

Modelling Financial Markets

Chapter 11: Continuous-Time Models



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How do we describe prices in continuous time?

What the models explain

- Can one model describe prices at **every** time scale at once?
  - prices are recorded daily or every few milliseconds
- Part I: Brownian motion, Itô calculus, simulation, geometric Brownian motion (GBM), mean reversion
- Part II: estimating diffusions, change of measure, jumps, stochastic volatility
  - then models against data, two case studies, AI-assisted discovery

What you should be able to do

- derive solutions with Itô's lemma and simulate a stochastic differential equation with a known error order
- compute GBM, Ornstein–Uhlenbeck, Merton and Heston estimates on data
  - with standard errors that respect what the sample identifies
- interpret which feature of the data each model reproduces, and which feature none of them does
- interpret why option pricing (Chapter 12) and interest-rate models use this language
- Further Topics and Appendix: reading for familiarity

Indices and Data Sources

- **VIX**: Cboe Volatility Index.
- **BET**: Bucharest Exchange Trading (index).
- **BVB**: Bursa de Valori București — the Bucharest Stock Exchange.
- **CBOE**: Chicago Board Options Exchange.
- **FRED**: Federal Reserve Economic Data.
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code).

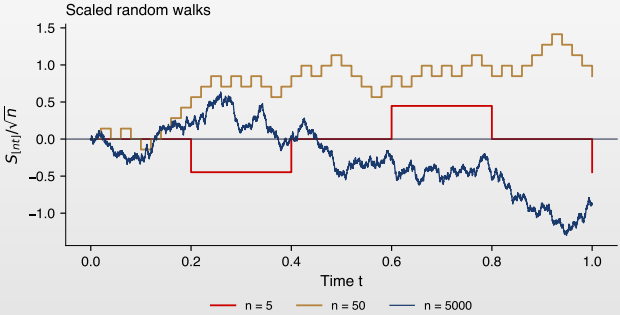
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

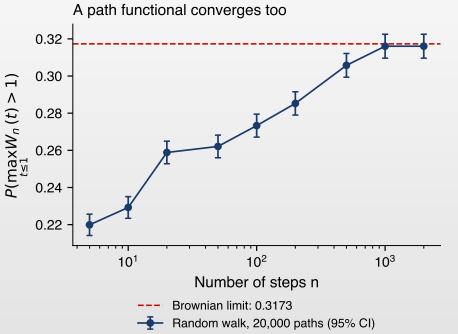
From Random Walk to Brownian Motion: Donsker's Theorem (1/3)

- **Question**: which process do we get as price steps become smaller and more frequent?
  - prices move in discrete steps
- **Random walk**: $S_0 = 0$, $S_k = S_{k-1} + \xi_k = \xi_1 + \cdots + \xi_k$
  - steps ξ_i i.i.d. (independent and identically distributed), mean 0 and variance 1
  - it is the log-price model of Chapter 2
  - ξ_k is the standardised return of day k
- **Scaled random walk**: $W_n(t) = S_{\lfloor nt \rfloor}/\sqrt{n}$, with n steps per unit of time
  - $\lfloor nt \rfloor$: integer part of nt
  - why $\sqrt{n}$: $\mathrm{Var}(W_n(t)) = \lfloor nt \rfloor / n \to t$
  - divided by n the limit would be 0
  - undivided the variance would explode
- Central limit theorem (CLT): $W_n(t) \to N(0, t)$ for each t separately
- Donsker's theorem: $W_n \Rightarrow W$ for the **whole path** Billingsley (1999)
  - $\Rightarrow$: convergence in distribution
  - holds whatever the law of the steps

From Random Walk to Brownian Motion: Donsker's Theorem (2/3)

CI: Confidence Interval.





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From Random Walk to Brownian Motion: Donsker's Theorem (3/3)

- Left: three scaled random walks on $[0, 1]$ with $n = 5$, 50 and 5 000 steps
  - the vertical axis is $S_{\lfloor nt \rfloor} / \sqrt{n}$
  - for large n the step path becomes continuous and irregular, like a Brownian path
- Right: $P(\max_{t \leq 1} W_n(t) > 1)$ against n on a log axis
  - estimated from 20 000 paths (95% CI)
  - the probability rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$)
  - the limit is the reflection-principle value $2(1 - \Phi(1)) = 0.3173$
  - Φ: standard Normal distribution function
- A coarse grid misses the maximum reached between observations
- Convergence is slow, of order $1/\sqrt{n}$
  - daily closes understate the intraday range

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Brownian Motion: Definition and Properties

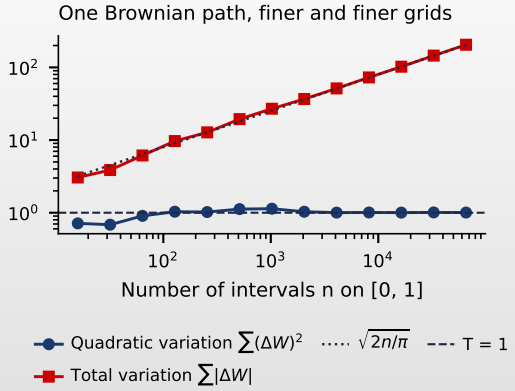
- **Standard Brownian motion**: four defining properties
  - $W_0 = 0$
  - independent increments
  - $W_t - W_s \sim N(0, t-s)$ for $s < t$
  - continuous paths
  - existence: Wiener (1923)
  - simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t}\, Z$, $Z \sim N(0,1)$
- A Gaussian process with $\mathrm{Cov}(W_s, W_t) = \min(s,t)$
- A **martingale**: $\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$
  - $\mathcal{F}_s$: the information up to s
  - the efficient-market random walk of Chapter 2
- **Self-similarity**: $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
  - $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \to \infty$
  - ordinary calculus fails

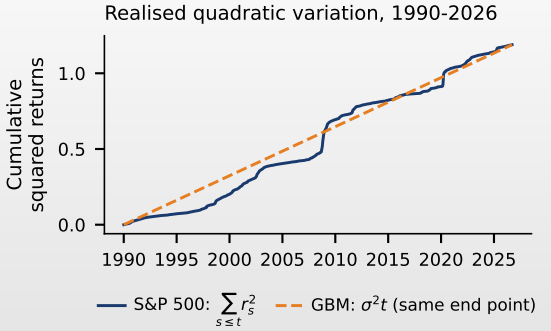
▸ Appendix: Brownian paths

Total and Quadratic Variation

- **Question**: why does ordinary calculus fail on a Brownian path, and what replaces it?
- **Total variation** $\sum_k |\Delta W_k|$
  - grid $0 = t_0 < \cdots < t_n = T$, step $\Delta t = T/n$, $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
  - its mean is $\sqrt{2nT/\pi} \to \infty$
  - a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
  - a random sum with a **deterministic** limit
  - $Q_n = \sum_{k=1}^{n} (\Delta W_k)^2$, $\mathrm{Var}(Q_n) = 2T^2/n$
  - one year of daily data gives about 9% relative error ($\sqrt{2/252}$)
  - the idea behind realised variance (Chapter 9)
- Shorthand of Itô calculus: $(\mathrm{d}W_t)^2 = \mathrm{d}t$

Quadratic Variation: Simulation and Data (1/2)





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Quadratic Variation: Simulation and Data (2/2)

- Left: one Brownian path on grids with 16 to 65,536 intervals
  - the quadratic variation settles at 1.006 ($T = 1$)
  - the total variation keeps growing: 205 against $\sqrt{2n/\pi} = 204$
- Right: cumulative $\sum r_t^2$ of the S&P 500, 1.188 over 36.7 years
  - average volatility 18.0%
  - under geometric Brownian motion it would follow the straight line $\sigma^2 t$
- The sum rises in steps
  - September 2008 – March 2009: 1.6% of the days, but 15.2% of the quadratic variation
  - February – April 2020: 0.5% of the days, but 7.8% of the quadratic variation
  - volatility is itself random (Heston, Section 10)

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The Itô Integral (1/2)

- Goal: a meaning for $\int_0^T H_t \, \mathrm{d}W_t$
  - the gain of holding H_t units of an asset priced W
  - Riemann–Stieltjes integration fails because the total variation is infinite
- **Itô integral**: the limit in mean square of sums with the integrand at the **left** end point
  - $\int_0^T H_t \, \mathrm{d}W_t = \lim_{n\to\infty} \sum_{k=1}^{n} H_{t_{k-1}} \left(W_{t_k} - W_{t_{k-1}}\right)$
  - for H **adapted** with $\mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right] < \infty$
  - H_t uses only information up to t
  - the position is chosen before the next price move

The Itô Integral (2/2)

- Zero mean: $\mathbb{E}\left[\int_0^T H_t \, \mathrm{d}W_t\right] = 0$
- **Itô isometry**: $\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right]$
- $t \mapsto \int_0^t H_s \, \mathrm{d}W_s$ is a martingale
  - no strategy earns money on average
  - weaker conditions: Appendix
- **Stratonovich** integral: the integrand averaged over the end points, $\frac{1}{2}(H_{t_{k-1}} + H_{t_k})$
  - it gives a different number
  - finance uses Itô, because that position would use the price at t_k

Appendix: worked Itô example

Itô Processes and Itô's Lemma

- **Itô process**: $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$
  - drift a_t, diffusion coefficient b_t, both adapted
  - shorthand for $X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}W_s$
- **Itô's lemma**: for $f(t,x)$ continuously differentiable once in t and twice in x Itô (1944)
  - $\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)\mathrm{d}t + b_t f_x\,\mathrm{d}W_t$
  - partial derivatives evaluated at (t, X_t)
  - $f_t = \partial f/\partial t$, $f_x = \partial f/\partial x$, $f_{xx} = \partial^2 f/\partial x^2$
- Rule of thumb: Taylor expansion to second order
  - with $\mathrm{d}t \cdot \mathrm{d}t = 0$, $\mathrm{d}t \cdot \mathrm{d}W = 0$, $\mathrm{d}W \cdot \mathrm{d}W = \mathrm{d}t$
  - the second-order term $\frac{1}{2}b_t^2 f_{xx}\,\mathrm{d}t$ does not vanish
  - this is the whole novelty

Appendix: quadratic covariation

Stochastic Differential Equations and the Euler–Maruyama Scheme

- **Question**: how do we simulate a model with no elementary, easy-to-simulate transition (Heston)?
  - exact simulation exists, but is costly: Broadie & Kaya (2006)
- **Question**: how large is the discretisation error?
- A **stochastic differential equation** (SDE): $\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$, X_0 given
  - drift and diffusion depend on the current state
  - GBM, Ornstein–Uhlenbeck and Merton have exact transitions
  - we simulate them with no discretisation error
  - Heston has only a closed-form characteristic function
  - Existence conditions: ▸ Appendix
- **Euler–Maruyama**: freeze a and b over each step Higham (2001)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k$, $\Delta W_k = \sqrt{\Delta t}\,Z_k$
  - X_k is the approximation at $t_k = k\Delta t$
  - Z_k i.i.d. $N(0,1)$

The Milstein Scheme and Strong Convergence

- **Milstein**: add the next Itô–Taylor term Milstein (1975)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k + \frac{1}{2} b(X_k)\,b'(X_k)\left[(\Delta W_k)^2 - \Delta t\right]$
  - $b' = \mathrm{d}b/\mathrm{d}x$
  - the correction has conditional mean zero
  - it raises the strong order (path accuracy), not the weak order
- **Strong error**: $\mathbb{E}|X_T - X_T^{\Delta t}| \le C\,\Delta t^{\gamma}$
  - how close each simulated path is to the true one
  - X_T is the exact solution, $X_T^{\Delta t}$ the scheme with step Δt
  - C is a constant and γ the strong order
  - $\gamma = 1/2$ for Euler–Maruyama and $\gamma = 1$ for Milstein
  - it matters for path-dependent quantities (hedging errors, barrier crossings)

Geometric Brownian Motion (GBM) from Itô's Lemma (1/2)

- **Question**: which price model stays positive?
  - it should give percentage returns with the same distribution at every price level
- SDE: $\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t$
  - the **percentage** change has constant drift μ and volatility σ per year
  - proposed by Osborne (1959) and Samuelson (1965)
  - the price model behind Black & Scholes (1973)
- Itô's lemma for $f(x) = \ln x$
  - $f_x = 1/x$, $f_{xx} = -1/x^2$, $a_t = \mu S_t$, $b_t = \sigma S_t$
  - $\mathrm{d} \ln S_t = \left(\mu - \frac{1}{2}\sigma^2\right) \mathrm{d}t + \sigma \,\mathrm{d}W_t$
  - so $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right)$
  - prices stay positive
  - exact simulation needs only Normal draws

Geometric Brownian Motion (GBM) from Itô's Lemma (2/2)

- The log price grows at the log drift $m = \mu - \sigma^2/2$, not at μ
  - volatility lowers the growth of the median investor's wealth
  - With $\sigma = 18\%$, the cost is $\sigma^2/2 = 1.6$ percentage points a year
  - With $\sigma = 67\%$ (Bitcoin), the cost reaches 22 percentage points a year
- Testable implications: $r_t = \ln(S_t/S_{t-\Delta t})$ i.i.d. $N(m\Delta t, \sigma^2\Delta t)$ at every frequency
  - Skewness and excess kurtosis are zero
  - There is no autocorrelation in r_t, $|r_t|$ or r_t^2
  - The variance is proportional to the horizon

▸ Appendix: lognormal outcomes

Estimating GBM by Maximum Likelihood (1/2)

- Maximum likelihood estimation (MLE) from n log returns at step Δt, over $T = n\Delta t$ years
  - $\hat{\sigma}^2 = \frac{1}{n\Delta t}\sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$ ($\bar{r}$: mean log return)
  - under GBM, the standard error (SE) $\mathrm{SE}(\hat{\sigma}) = \sigma/\sqrt{2n}$ shrinks with the number of observations
  - $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$ depends only on the **calendar span**
  - reason: $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

	Years	$\hat{\sigma}$ (%)	$\mathrm{SE}(\hat{\sigma})$	$\hat{m}$ (%)	$\mathrm{SE}(\hat{m})$
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

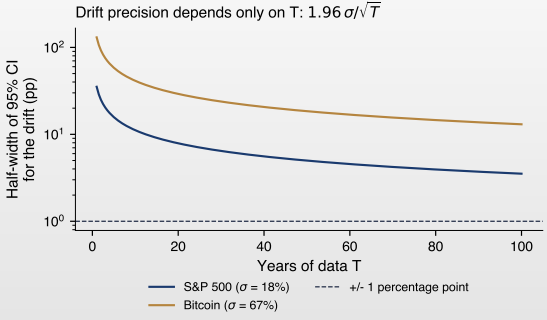
- The SE is 0.1–0.7 percentage points for volatility, but 3–19 points for the log drift
- S&P 500: $\hat{\mu} = 8.3\% + 1.62\% = 10.0\%$ per year

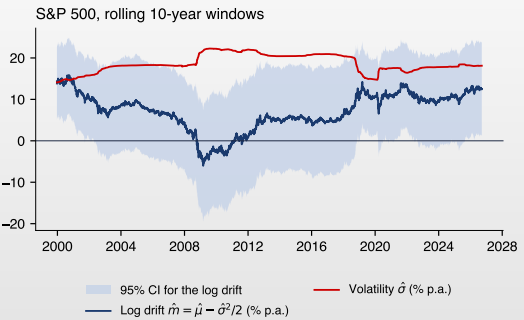
Estimating GBM by Maximum Likelihood (2/2)

- These SE are conditional on GBM
  - with excess kurtosis k, even independent returns inflate $\mathrm{SE}(\hat{\sigma})$ by $\sqrt{(k+2)/2} = 2.5$ (S&P 500)
  - block-bootstrap SE: Seminar 11, B1
  - the drift SE needs only a finite variance
  - the volatility SE requires a finite fourth moment, which Chapter 1 questions

How Precisely Can We Estimate the Drift? (1/2)

p.p. = percentage points.





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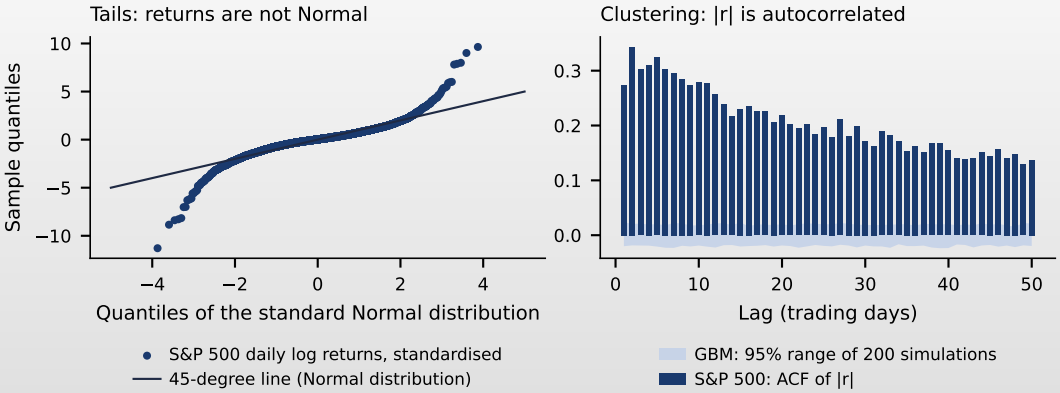
How Precisely Can We Estimate the Drift? (2/2)

- Left: half-width of the 95% CI for the drift against years of data
- A precision of ± 1 point needs $(1.96\,\hat{\sigma}/0.01)^2$ years of data
  - S&P 500: $(1.96\,\hat{\sigma}/0.01)^2 = 1\,243$ years
  - 17 028 years for Bitcoin
- Right: S&P 500 on rolling 10-year windows
  - the log drift ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09)
  - each estimate has a CI of about ± 11 points
  - with this precision, the swings cannot be safely attributed to changes in the drift
  - volatility stays between 14.1% and 22.3%: precise, but not constant
- Option pricing in Chapter 12 avoids μ
  - under the risk-neutral measure only σ enters (Girsanov, Section 8)

MFM_ch11_drift_precision

Does GBM Fit the S&P 500? (1/2)

ACF: AutoCorrelation Function.



MFM_ch11_gbm_vs_data ▶ Interactive

Does GBM Fit the S&P 500? (2/2)

- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution
  - excess kurtosis 10.9 instead of 0
  - the sample kurtosis consistently estimates a finite kurtosis only with a finite fourth moment, which Chapter 1 questions
  - inference on the kurtosis usually requires a finite eighth moment
  - worst day at -11.3 standard deviations
  - 62 days beyond ± 4 standard deviations, against 0.59 expected under the Normal distribution
  - Jarque–Bera (JB) statistic: 45 805 against the 5% critical value 5.99 (χ^2_2)
  - JB tests zero skewness and zero excess kurtosis jointly
- Right: autocorrelation function (ACF) of $|r_t|$
  - 0.27 at lag 1 and still 0.14 at lag 50
  - the 95% band from 200 GBM samples of the same length stays below 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 Cont (2001)
  - the rest of the chapter adds mean reversion, jumps and random volatility
- What the result supports: description
  - the data reject two implications of GBM without identifying the mechanism that produces them

MFM_ch11_gbm_vs_data

The Ornstein–Uhlenbeck (OU) Process

- **Question**: how do we model a variable that returns to a normal level?
  - examples: an interest rate or the VIX
  - unlike a stock price, it does not drift away
- SDE: $\mathrm{d}x_t = \kappa(\theta - x_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930)
  - θ: long-run mean; κ: speed of mean reversion; σ: volatility
  - Above θ the drift is negative, below θ positive
  - the drift is a force pulling x back to θ
  - Used for interest rates Vasicek (1977), volatility, spreads and commodity prices
- **Half-life** $\ln 2/\kappa$: the time for the expected deviation from θ to halve
  - With $\kappa = 0.5$ per year, the half-life is 1.39 years

▸ Appendix: solving the OU equation

Exact Discretisation: an AR(1)

- Sampled at step Δt, the OU process is exactly an autoregressive model of order 1 (AR(1))
  - $x_{t+\Delta t} = a + b\,x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1-b)$, $\mathrm{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1-b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t
  - OLS gives the intercept $\hat{a}$, the slope $\hat{b}$ and the residual standard deviation $\hat{s}_\varepsilon$
  - then invert the mapping: $\hat{\kappa} = -\ln \hat{b}/\Delta t$, $\hat{\theta} = \hat{a}/(1-\hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa}/(1-\hat{b}^2)}$
- Standard errors of $\hat{\kappa}$ and $\hat{\theta}$: the delta method, from the OLS covariance of $(\hat{a}, \hat{b})$
  - $\hat{\kappa}$ depends on $\hat{b}$ only: $\partial\kappa/\partial b = -1/(b\Delta t)$
  - so $\mathrm{Var}(\hat{b})$ suffices (Seminar 11, A1)
- With daily data b is very close to 1
  - κ is estimated imprecisely, as in unit-root tests

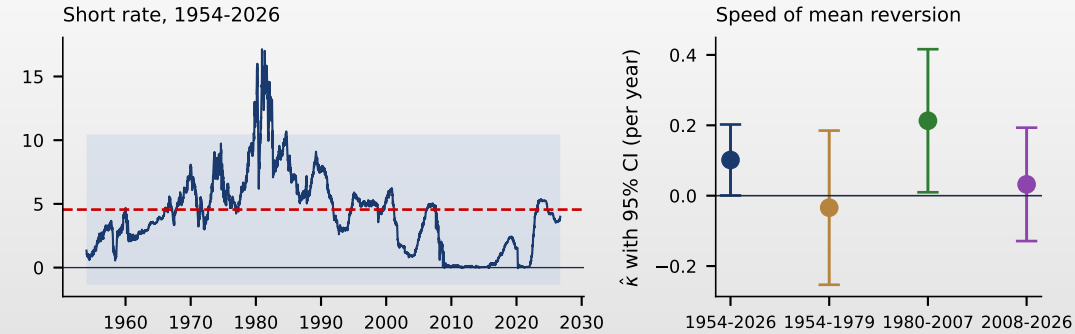
▸ Appendix: three speeds of reversion

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process Vasicek (1977)
  - Normal distribution: negative rates have positive probability
  - the long-run probability is $\Phi(-\theta\sqrt{2\kappa}/\sigma)$
  - Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross** (CIR): $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t$ Cox, Ingersoll & Ross (1985)
  - Volatility shrinks near zero
  - The rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) Feller (1951)
- Which is closer to the data is an empirical question Chan, Karolyi, Longstaff & Sanders (1992)
  - tested in Section 7

Vasicek on the 3-Month Treasury Bill Rate (1/2)

s.d. = standard deviation.



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Vasicek on the 3-Month Treasury Bill Rate (2/2)

- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026
  - with the Vasicek long-run mean $\hat{\theta} = 4.55\%$
  - band: $\hat{\theta}$ plus/minus two stationary standard deviations (s.d.)
  - $\hat{\kappa} = 0.101$ (SE 0.051), a half-life of 6.8 years
  - volatility $\hat{\sigma} = 1.33\%$
  - Daily AR(1) coefficient $\hat{b} = 0.99960$: almost a unit root
- Right: $\hat{\kappa}$ with 95% CI over three periods
  - -0.034 in 1954–1979 (explosive), 0.213 in 1980–2007 and 0.032 in 2008–2026
  - Two of the three CIs include zero
- Test of the random-walk null ($\kappa = 0$): parametric bootstrap
  - 500 random walks of the same length
  - reason: under $\kappa = 0$ the distribution of $\hat{\kappa}$ is not Normal and is shifted upwards
  - p-value $= 0.287$: the sample does not establish mean reversion (Seminar 11, B2)
- What the result supports: description

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Infill Asymptotics: What Sampling More Often Buys

Question: which parameters do we learn from sampling more often, and which only from more years?

Infill: $\Delta \to 0$ on a fixed $[0, T]$

- $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$
- Quadratic variation, hence b_t^2 along the path (under regularity): identified without error in the limit
- The whole diffusion function or its parameters need further conditions
  - Quadratic variation is observable (continuous Itô process, no measurement noise)
- Sampling more often helps σ and $b(x)$
  - σ: Section 5; $b(x)$: nonparametric, Further Topics

Long-Span Asymptotics: What Only More Years Buy

Long span: $T \to \infty$

- The drift is not consistently estimable on a fixed span
  - Fisher information $I(m) = -\mathbb{E}[\partial^2 \ell / \partial m^2]$ (ℓ: log-likelihood) bounds $\mathrm{Var}(\hat{m}) \geq 1/I(m)$
  - for n GBM returns $I(m) = n\Delta t/\sigma^2 = T/\sigma^2$
  - so $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$
- OU: $\mathrm{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \to 2\kappa/T$ as $\Delta \to 0$
  - derivation: Seminar 11, A1
- The bias of $\hat{\kappa}$ does not shrink with finer sampling Yu (2012)
  - $b = e^{-\kappa\Delta} \to 1$
  - $\hat{\kappa}$ is far from Normal when κT is small

- The imprecise GBM drift and the biased $\hat{\kappa}$ have the same cause
  - information grows with T, not with n

Why Prices Use a Different Probability: a One-Period Example (1/2)

- A stock costs 100 today and tomorrow is worth 120 (up) or 90 (down)
  - the interest rate is 0
  - the real-world probability of up is $p = 0.6$
- A claim pays 1 if up and 0 if down
  - replicate it with Δ shares and B in cash
  - $120\Delta + B = 1$ and $90\Delta + B = 0$ give $\Delta = 1/30$, $B = -3$
  - the replica costs $1/3$, and any other price is an arbitrage
- The same price is the expectation under $q = (100 - 90)/(120 - 90) = 1/3$
  - q is the only weight with $120q + 90(1 - q) = 100$
  - p never enters

Why Prices Use a Different Probability: a One-Period Example (2/2)

- In general, absence of arbitrage means that discounted traded prices are martingales under an equivalent measure Q
  - fundamental theorem of asset pricing, Shreve (2004), Ch. 5
  - Girsanov builds Q in continuous time
- Check: what is the price of the claim if $p = 0.9$?
- And if the down state is 80 instead of 90?
  - Answer: still 1/3
  - with 80, $q = 20/40 = 1/2$, so the price is 1/2

Girsanov's Theorem (1/2)

- Real-world measure P and an equivalent measure Q, linked by the **Radon–Nikodym density**
  - equivalent: the same events of probability zero
  - $\left.\frac{\mathrm{d}Q}{\mathrm{d}P}\right|_{\mathcal{F}_T} = Z_T = \exp\left(-\int_0^T \theta_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^T \theta_s^2 \,\mathrm{d}s\right)$
  - θ_s is adapted (below: the market price of risk)
  - $Z_T > 0$ with $\mathbb{E}^P[Z_T] = 1$ reweights the probability of each path
  - $Q(A) = \mathbb{E}^P[Z_T 1_A]$
- Girsanov (1960): under Q, $\tilde{W}_t = W_t + \int_0^t \theta_s \,\mathrm{d}s$ is a Brownian motion
  - condition: $\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \theta_s^2 \mathrm{d}s}\right] < \infty$ (Novikov condition)

Girsanov's Theorem (2/2)

- For $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$: $\mathrm{d}X = (a - b\theta)\,\mathrm{d}t + b\,\mathrm{d}\tilde{W}$
- For continuous Itô processes **only the drift changes**
  - the diffusion and the quadratic variation are the same under P and Q
- With jumps (Section 9), Q may also change the jump intensity and the jump-size distribution
  - $\hat{\lambda}$ from returns is not the intensity that prices options
- With stochastic volatility, P and Q weigh future variance paths differently
  - historical variance is a benchmark, not a calibration

Market Price of Risk: P versus Q (1/2)

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**
  - under Q, $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}\tilde{W}$
  - $e^{-rt}S_t$ is a Q-martingale (Seminar 11, A4)
- S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$
  - r: the mean 3-month T-bill rate
  - SE $1/\sqrt{T} = 0.17$ under GBM, as imprecise as the drift
  - beyond GBM: the delta method, with the long-run covariance of the first and second moments of returns
  - this requires a finite fourth moment, which Chapter 1 questions

Market Price of Risk: P versus Q (2/2)

- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$
  - θ, θ^Q: long-run levels
  - The long-run level that prices bonds differs from the one estimated on data (Section 6)
  - A positive term premium means $\lambda < 0$
  - then $\theta^Q = \theta - \lambda\sigma/\kappa > \theta$ and long yields are higher
- Heston with volatility risk premium λv_t: $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
  - this λ is a separate parameter
  - $\lambda < 0$: investors pay for protection against variance
  - it gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE (1/2)

- If $\mathrm{d}X = a(X)\,\mathrm{d}t + b(X)\,\mathrm{d}\tilde{W}$ under Q Kac (1949)
  - $u(t,x) = \mathbb{E}^Q\left[e^{-\int_t^T r_s\,\mathrm{d}s}\, g(X_T) \mid X_t = x\right]$ solves a PDE (partial differential equation)
  - the PDE: $u_t + a\,u_x + \frac{1}{2}b^2 u_{xx} - r\,u = 0$, $u(T,x) = g(x)$
  - u is the price at t of the pay-off $g(X_T)$ at T
  - the pay-off is discounted at the short rate r_s
- Proof idea: $e^{-\int_0^t r_s\mathrm{d}s}\,u(t,X_t)$ is a Q-martingale
  - so its Itô drift is zero
  - by the same route, Chapter 12 derives Black–Scholes

Feynman–Kac: From Expectations to the Pricing PDE (2/2)

- Vasicek bond ($g = 1$, $X = r$): $p(t, r; T) = \exp\left(A(\tau) - B(\tau)\, r\right)$, $\tau = T - t$
  - derivation: Seminar 11, A6
  - $B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}$
  - $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2}\right)(B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
- Yields $y(\tau) = (B(\tau)\, r - A(\tau))/\tau$ tend to $\theta^Q - \sigma^2/(2\kappa^2)$
  - rate uncertainty lowers long yields (convexity)
  - comparison with Treasury yields: ▸ Appendix

Recap: Pricing under Q

- Formulas to keep
  - Girsanov: under Q the drift a becomes $a - b\theta$, the diffusion b is unchanged
  - GBM: $\theta = (\mu - r)/\sigma$ turns the drift μ into r
  - so $e^{-rt}S_t$ is a Q-martingale
  - Feynman–Kac: price $= \mathbb{E}^Q$[discounted pay-off], the solution of the pricing PDE
- Check: what are the market price of risk and the drift of S under Q?
  - inputs: $\mu = 9\%$, $\sigma = 20\%$, $r = 3\%$
  - Answer: $\theta = 0.06/0.20 = 0.30$
  - drift under Q: $\mu - \sigma\theta = 9\% - 6\% = 3\% = r$
  - $\sigma = 20\%$ is unchanged

Merton's Jump-Diffusion (1/2)

- **Question**: which model explains crash days that are practically impossible under GBM?
- Brownian paths are continuous, but prices gap on news
  - on 19 October 1987 the Dow Jones fell 22.6%
  - about 20 standard deviations under GBM with 18% volatility
  - A jump cannot be hedged by continuous trading Merton (1976)
- Log price = Brownian motion with drift + **compound Poisson** jumps
  - $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$, $J_t = \sum_{i \le N_t} Y_i$
  - N_t is a Poisson process with intensity λ
  - the jump sizes $Y_i \sim N(\mu_J, \sigma_J^2)$ are i.i.d.
  - W, N and Y are independent
  - the model has five parameters, $(m, \sigma, \lambda, \mu_J, \sigma_J)$

Merton's Jump-Diffusion (2/2)

- Given k jumps, the return over Δt is $N\left(m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$
- Unconditionally, it is a **Poisson mixture** of Normal densities, i.i.d. over time
  - $f(r) = \sum_{k\geq 0} e^{-\lambda\Delta t}\frac{(\lambda\Delta t)^k}{k!}\,\varphi\left(r;\ m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$
  - Weight: Poisson probability of k jumps
  - $\varphi(r;\text{mean},\text{variance})$: the Normal density
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$
  - then draw from the Normal distribution with the conditional moments

Appendix: Merton moments

Merton Estimates for the S&P 500, BET and Bitcoin (1/2)

LR (Likelihood Ratio): the likelihood-ratio statistic.

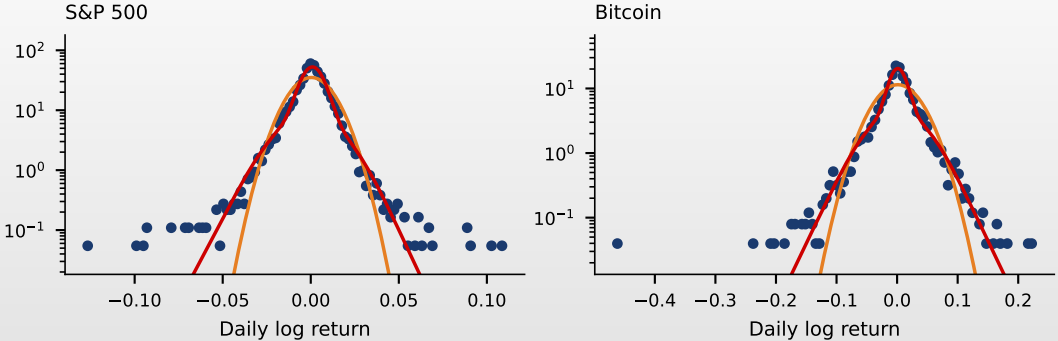
	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	−0.16	1.44	71%	2 401
BET	11.3	79 (7)	−0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- The Poisson-mixture log-likelihood is maximised numerically
  - up to 12 jumps per day
  - in brackets, SE from the Hessian: conditional on the i.i.d. model, around the local maximum
  - sandwich SE of $\hat{\lambda}$ (scores with a HAC covariance): 28 (S&P 500), 12 (BET), 36 (Bitcoin)
- The likelihood is **unbounded**, so the MLE is a local maximum Honoré (1998)
  - with $m\Delta t$ equal to one observed return and $\sigma \to 0$ that density explodes
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Estimates for the S&P 500, BET and Bitcoin (2/2)

- The LR test against GBM ($\lambda = 0$) is defined on $\sigma \geq 5\%$
- The LR statistic is not χ^2_3
  - $\lambda = 0$ lies on the boundary
  - μ_J, σ_J are unidentified
  - bootstrap critical values: Seminar 11, B5; the test is about the presence of jumps, not the precision of the parameters

Merton Fit to the Return Distribution (1/2)



MFM_ch11_merton_density ▶ Interactive

Merton Fit to the Return Distribution (2/2)

- Density of daily log returns on a log scale
  - histogram, the Normal distribution implied by GBM and the fitted Merton mixture
  - Merton follows the peak and the shoulders far better
- S&P 500: Merton versus data
  - excess kurtosis: 3.7 against 10.9 in the data
  - skewness: -0.31 against -0.36
- The most extreme days lie beyond both fitted curves
- About 102 "jumps" a year
  - the model uses jumps to mimic **changing volatility**
- An i.i.d. model cannot tell a jump from a volatile period
  - so a direct test is needed
- What the result supports: description

MFM_ch11_merton_density

▸ Appendix: detecting jumps

Recap: What Jumps Add

- Formulas to keep
  - $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$: mean log return $m + \lambda\mu_J$ per year
  - so m alone is the drift of the diffusion part only
  - Higher cumulants come only from the jumps: $c_k = \lambda\Delta t\,\mathbb{E}[Y^k]$ for $k \geq 3$
  - Excess kurtosis c_4/c_2^2 falls like $1/\Delta t$ with the horizon
- From the data: S&P 500 $\hat{m} + \hat{\lambda}\hat{\mu}_J = 24.8\% + (-16.5\%) \approx 8.3\%$
  - the same as the GBM log drift
- Check: jumps give a daily excess kurtosis of 12
- What is it for 10-day returns, if returns are i.i.d.?
  - Answer: $12/10 = 1.2$
  - over 10 days both c_4 and c_2 grow 10 times, so c_4/c_2^2 falls 10 times

The Heston Model (1/2)

- **Question**: which model generates volatility clustering and negative skewness?
  - here, negative skewness generated by the leverage effect
  - GBM and Merton with constant parameters produce neither volatility clustering nor return–volatility dependence
  - Merton can produce negative marginal skewness, through jumps with $\mu_J < 0$
- Two SDEs, for the log price and for the instantaneous variance v_t Heston (1993)
  - $\mathrm{d} \ln S_t = \left(\mu - \frac{1}{2} v_t\right) \mathrm{d}t + \sqrt{v_t}\, \mathrm{d}W_t^{(1)}$
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\, \mathrm{d}t + \xi \sqrt{v_t}\, \mathrm{d}W_t^{(2)}$, $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$
  - the variance is a CIR process: level θ, speed κ, volatility of volatility ξ
  - Feller condition $2\kappa\theta \geq \xi^2$: the variance never touches zero Feller (1951)

The Heston Model (2/2)

- $\rho < 0$: the **leverage effect** (Section 11)
  - a price fall raises the variance that scales the next moves
  - so returns are negatively skewed
- Simulation: two schemes
  - Euler scheme with full truncation $v^+ = \max(v, 0)$ Lord, Koekkoek & van Dijk (2010)
  - or the quadratic-exponential scheme Andersen (2008)
- Daily returns need several steps per day
  - with one step, the daily return is conditionally Normal given v at the start of the day
  - one step removes the intraday channel through which ρ shapes skewness; the unconditional skewness can be small, but not necessarily zero
  - Section 11 uses 390 steps per day, one per minute of a 6.5-hour trading session

Heston Parameters from the VIX

- Proxy $y_t = (\mathrm{VIX}_t/100)^2$ in place of v_t
  - discretised CIR regression without an extra intercept
  - $\frac{y_{t+\Delta t}-y_t}{\sqrt{y_t}} = \kappa\theta\frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\,\varepsilon_t$
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$, $\hat{\xi} = 0.56$ and $\hat{\rho} = -0.69$
  - half-life: 42 days
  - The long-run level from the VIX is $\sqrt{\hat{\theta}} = 20.9\%$
  - the realised volatility is 18.0%
  - The Feller ratio of the proxy is $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t, so $\hat{\xi}$ is attenuated (Further Topics)
  - y_t is a risk-neutral expectation of future variance, affine in v_t
  - For the simulated paths we set θ to the realised variance of the S&P 500

Recap: Stochastic Volatility

- Formulas to keep
  - Variance is a CIR process, $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$
  - it stays positive if $2\kappa\theta \geq \xi^2$
  - $\rho = \mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) < 0$ gives leverage and negatively skewed returns
  - it also gives a downward implied-volatility skew
  - $(\mathrm{VIX}/100)^2 = a + b\,v_t$ with Q-parameters: κ and ρ carry over
  - ξ is attenuated by b
- Check: with $\kappa = 4$ and $\theta = 0.04$, what is the largest ξ for which the Feller condition holds?
  - Answer: $\xi \leq \sqrt{2 \cdot 4 \cdot 0.04} = \sqrt{0.32} \approx 0.57$
  - a larger vol-of-vol lets the variance touch zero

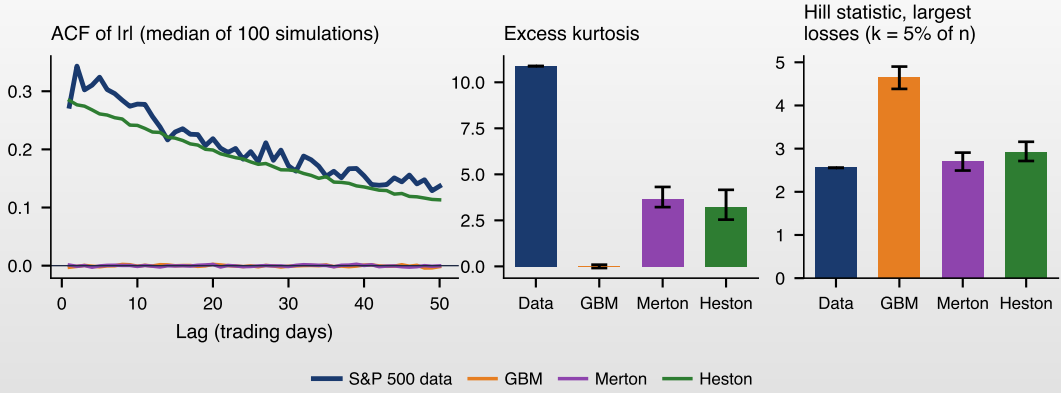
Three Yardsticks for Simulated Returns (1/2)

- **Question**: which estimated model (GBM, Merton, Heston) reproduces S&P 500 returns?
  - three features: the shape, the volatility clustering and the tail
- **Shape**: skewness $c_3/c_2^{3/2}$ and excess kurtosis c_4/c_2^2 of daily returns
  - cumulants: Further Topics
  - the kurtosis yardstick estimates a finite kurtosis consistently only with a finite fourth moment, which Chapter 1 questions
- **Clustering**: the ACF of $|r_t|$ at lags 1–20
  - zero for any i.i.d. model, Merton included
- **Tail**: Hill statistic of the k largest losses $L_{(1)} \geq \cdots \geq L_{(k+1)}$
  - $\hat{\alpha} = \left[\frac{1}{k}\sum_{i=1}^{k} \ln\left(L_{(i)}/L_{(k+1)}\right)\right]^{-1}$

Three Yardsticks for Simulated Returns (2/2)

- The Hill statistic estimates α in $P(L > x) \approx Cx^{-\alpha}$ for large x
  - $P(L > 2x)/P(L > x) \approx 2^{-\alpha}$, so a lower α is a heavier tail
- With $\alpha = 2.56$, doubling the loss leaves $2^{-2.56} \approx 0.17$ of the frequency
  - a Normal tail leaves $P(Z > 4)/P(Z > 2) \approx 0.0014$
- The estimate depends on k
  - for the data: 2.97, 2.56 and 2.04 at $k = 2.5\%$, 5% and 10% of the returns

Which Model Reproduces the S&P 500? (1/2)



MFM_ch11_models_vs_data ▶ Interactive

Which Model Reproduces the S&P 500? (2/2)

- 100 simulated samples of 9 245 days from each fitted model (medians, 90% ranges)
  - statistics compared: ACF of $|r_t|$, excess kurtosis (10.88 in the data)
  - and the Hill statistic at $k = 5\%$ of the returns
- Merton gets the tail and the skewness, but no clustering
  - Hill statistic: 2.70 against 2.56
  - skewness: -0.32 against -0.36
- Heston, with the VIX-based parameters, gets the clustering and part of the skewness
  - skewness: -0.21 against -0.36
  - the leading-order approximation gives about -0.20 at these parameters (Further Topics)
  - simulation: 390 full-truncation Euler steps per day, summed to daily returns
- No fitted model reaches the kurtosis of the data (full table: Appendix)
  - Andersen, Benzoni & Lund (2002) and Eraker, Johannes & Polson (2003) favour jumps **and** stochastic volatility
  - SVJ: stochastic volatility with jumps, Bates (1996)
- What the result supports: comparison of procedures
  - the models at their estimated parameters: a Heston model with a lower Feller ratio could reach the kurtosis (Appendix)

MFM_ch11_models_vs_data

Which Model When? Features, Identification and Limits

Feature in the data	Model	What the data identify	Unresolved
Positive prices, $\sqrt{t}$ scaling	GBM	σ precisely; μ only over very long spans	fat tails, clustering
Mean reversion of rates and of the VIX	OU, Vasicek, CIR, CKLS	$b(r)$ precisely; κ biased upwards, needs a long span	one factor too simple for the VIX
Gaps, short-horizon fat tails	Merton	λ, σ_J only under i.i.d. returns; the jump share depends on the frequency	no clustering
Clustering, leverage, implied-volatility skew	Heston	κ, ρ from the VIX; ξ attenuated; $\theta^Q \neq \theta^P$	kurtosis of the data
All of the above	Bates (Heston + jumps)	returns and option surfaces jointly	latent states: filtering needed

- Roughness and long memory of volatility (next): none of these one-factor models produces them

Background: Fractional Brownian Motion and the Hurst Exponent (deeper) (1/2)

- **Question**: can one process describe volatility that is both irregular and persistent?
  - irregular at small scales, persistent at large scales (the VIX ACF, Section 6)
- **Fractional Brownian motion** B^H, $0 < H < 1$: a Gaussian process with $B_0^H = 0$
  - $\mathrm{Cov}(B_s^H, B_t^H) = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$ gives $\min(s,t)$, i.e. Brownian motion
  - increments have variance $\mathbb{E}[(B_{t+h}^H - B_t^H)^2] = h^{2H}$
  - $H < 1/2$: successive increments are negatively correlated, paths rougher than Brownian motion
  - $H > 1/2$: positively correlated, smoother

Background: Fractional Brownian Motion and the Hurst Exponent (deeper) (2/2)

- **Fractional OU**: the OU equation driven by B^H instead of W
  - one H sets both ends of its ACF
- Short lags: the variogram behaves like $h^{2H} = h^{2\alpha+1}$
  - the roughness index is $\alpha = H - 1/2$
  - $\alpha < 0$: rough
- Long lags: the ACF decays like $h^{2H-2} = h^{-\beta}$
  - so $\beta = 2 - 2H$
  - $\beta < 1$: non-integrable ACF, long memory

Research Question: Rough, Yet Persistent?

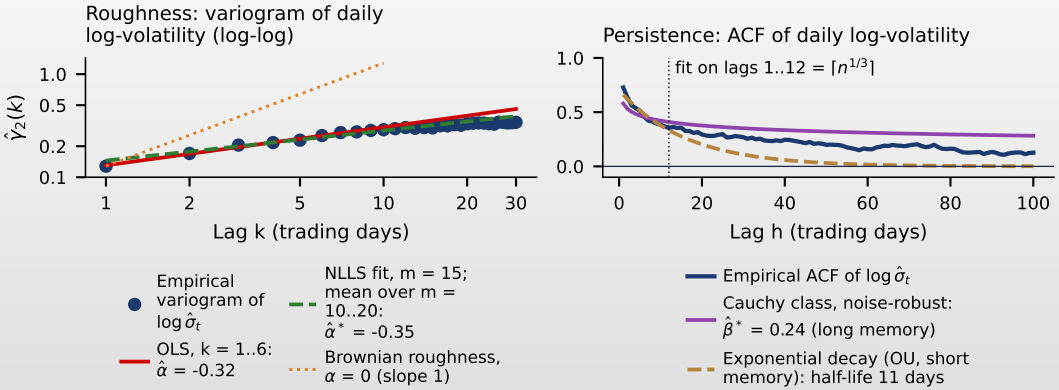
- Check: which H gives $\beta = 0.18$? Is that process rough?
  - Answer: $H = 1 - \beta/2 = 0.91$, so $\alpha = 0.41 > 0$: smooth, not rough
  - one H cannot match both features
- Paper: Bennedsen, Lunde & Pakkanen (2022), *Journal of Financial Econometrics*
  - it treats α and β as separate parameters
  - builds on Gatheral, Jaisson & Rosenbaum (2018): log-volatility as a fractional OU process with $H < 1/2$
  - RFSV: rough fractional stochastic volatility
  - The long-memory model of Comte & Renault (1998) uses $H > 1/2$
- Question 1: are realised volatilities both rough and persistent?
- Question 2: do models that set α and β separately forecast better?
  - model, estimators and results: ▸ Appendix

Case Study: Replication on SPY, $\Delta = 1$ Day (1/3)

- Data: SPY 5-minute bars, 1 468 full days (October 2020 – September 2026)
  - the paper studies E-mini S&P 500 futures; SPY tracks the same index
- $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log \mathrm{BV}_t$
  - BV_t: the daily bipower variation of the 78 five-minute returns
  - robust to jumps Barndorff-Nielsen & Shephard (2004)
- Roughness: variogram $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_t (x_{t+k} - x_t)^2 \approx c\, k^{2\alpha+1}$ for small k
  - OLS: the slope $\hat{a}_1$ of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \ldots, 6$
  - it gives $\hat{\alpha} = (\hat{a}_1 - 1)/2$
  - noise-robust NLLS: $\hat{\gamma}_2(k) = b_0 + b_1 k^{2\alpha+1}$
  - b_0 absorbs the measurement noise in x_t
- Persistence: Cauchy-class ACF $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$
  - fitted on the first $\lceil n^{1/3} \rceil = 12$ lags
- Short-memory benchmark: an OU process, with ACF $e^{-\kappa h}$

Case Study: Replication on SPY, $\Delta = 1$ Day (2/3)

NLLS: Non-Linear Least Squares.



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Case Study: Replication on SPY, $\Delta = 1$ Day (3/3)

- Left, roughness: empirical variogram $\hat{\gamma}_2(k)$ against k on log axes
  - the dotted line with slope 1 is Brownian roughness ($\alpha = 0$)
  - OLS gives $\hat{\alpha}_{\mathrm{OLS}} = -0.32$
  - the noise-robust NLLS gives $\hat{\alpha}^*_{\mathrm{NLLS}} = -0.35$
  - Bennedsen, Lunde & Pakkanen (2022) at 1 day: -0.30 (OLS) and -0.33 (NLLS)
- Right, persistence: the autocorrelation is 0.13 at lag 100
  - The Cauchy fit on 12 lags gives $\hat{\beta} = 0.26$ and $\hat{\beta}^* = 0.24$ (noise-robust)
  - both below 1: consistent with long memory, not a proof
  - 5-minute bars without pre-averaging; realised measures estimate integrated, not spot, variance
  - An OU fit (half-life 11 days) implies an autocorrelation of 0.002 at lag 100
- In the plot, m is the number of lags used in NLLS
  - 15 for the illustrated fit
  - the reported estimate averages fits over 10–20 lags
- What the result supports: description

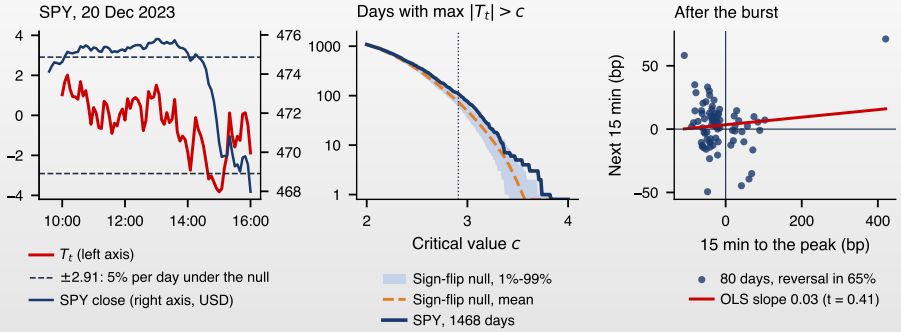
MFM_ch11_rough_vol

▸ Appendix: comparison with Table 3

Second Case Study (1/6): Drift Bursts

- Paper: Christensen, Oomen & Renò (2022), *Journal of Econometrics*, 227(2), 461–497
  - Data: quotes of six CME futures contracts, 2012–2017 (E-mini S&P 500 from 2010)
  - the contracts: E-mini S&P 500, 10-year T-Note, euro, gold, crude oil, corn
- Question: can the drift μ_t dominate volatility for a few minutes?
  - over long spans the drift is imprecise: ± 11 percentage points over 10 years (Section 5)
  - drift burst: before a time τ, $|\mu_t|$ grows like $(\tau - t)^{-a}$, with $1/2 < a < 1$
  - volatility may rise too, but more slowly; the price stays continuous, with no jump
  - the flash crashes of 6 May 2010 (equities) and 15 October 2014 (US Treasuries) are the paper's examples
- Main claim (the paper's abstract)
  - drift bursts occur once a week on average, in all four asset classes
  - most of them are followed by a price reversal
  - the reversal is stronger after negative drift bursts with large volume

Second Case Study (2/6): Drift Bursts in SPY, 5-Minute Bars



- **Left: the SPY price and the statistic T_t on 20 December 2023, the day with the largest $|T_t|$ in the sample (-3.83 at 15:00)**
- **Middle: the number of days with $\max_t |T_t| > c$, in the data and under the random-sign null (no drift)**
- **Right: the return in the 15 minutes after the peak of $|T_t|$ against the return in the 15 minutes before it, on days with $\max_t |T_t| > 2.91$**

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Second Case Study (3/6): The Paper's Statistic and the Course Adaptation

- The paper's statistic (Eq. 11–14), with the kernel $K(x) = e^{-|x|}$ for $x \leq 0$ (past only)
  - $\hat{\mu}_t = \frac{1}{h} \sum_i K\left(\frac{t_i - \mathbf{1} - t}{h}\right) \Delta_i X$ and $\hat{\sigma}_t^{\mathbf{2}} = \frac{1}{h'} \sum_i K\left(\frac{t_i - \mathbf{1} - t}{h'}\right) (\Delta_i X)^{\mathbf{2}}$
  - $T_t = \sqrt{h/K_{\mathbf{2}}}\, \hat{\mu}_t / \hat{\sigma}_t$, with $K_{\mathbf{2}} = \int K^{\mathbf{2}} = 1/2$
  - without a drift burst, T_t is asymptotically standard Normal; with one, $|T_t| \to \infty$
- The paper: $h = 5$ minutes, $h' = 25$ minutes, pre-averaging and a HAC (heteroskedasticity and autocorrelation consistent) correction for microstructure noise
  - the paper counts drift bursts at fixed thresholds, 4–6, and estimates the false alarms by simulation
- The course: an adaptation to five-minute bars, with the sums starting at the open
  - $\hat{s}_j^{\mathbf{2}} = \dfrac{\sum_i u_{ij} r_i^{\mathbf{2}}}{\Delta \sum_i u_{ij}}$ and $T_j = \dfrac{\sum_i w_{ij} r_i}{\sqrt{\hat{s}_j^{\mathbf{2}}\, \Delta \sum_i w_{ij}^{\mathbf{2}}}}$
  - the weights: $w_{ij} = e^{-(t_j - t_i - \mathbf{1})/h}$ and $u_{ij} = e^{-(t_j - t_i - \mathbf{1})/h'}$
  - the denominator is random and computed from the same returns, so T_j is not exactly Normal
  - with $h/\Delta = 3$ bars the asymptotic approximation is not guaranteed: critical values are calibrated by simulation

Second Case Study (4/6): The SPY Replication

- SPY, 1 468 complete days (October 2020 – September 2026)
  - $h = 15$ minutes (3 bars), $h' = 75$ minutes: the paper's 1:5 ratio
  - 73 tests a day, 10:00–16:00
  - no pre-averaging and no HAC: noise is small at 5 minutes
- The null: returns multiplied by random signs ± 1, 200 replications
  - it keeps the absolute returns and the intraday pattern, but removes the drift
  - it also removes jump asymmetry, sign dependence and the leverage effect
  - so it is narrower than "no drift burst"; the paper uses a jump-diffusion null
- Critical values
  - 2.91: the 95% quantile of $\max_t |T_t|$ under the null
  - Gumbel: 3.50, exceeded on only 0.12% of the simulated days
  - a single jump gives at most $|T_t| = 2.74$ when the jump return dominates both kernel sums

Second Case Study (5/6): The Result

- 109 candidate days with $\max_t |T_t| > 2.91$, against 73.84 on average under the null
  - none of the 200 replications reaches 109 (maximum 100): $p_{\mathrm{MC}} = 0.005$, under the random-sign null
  - at $c = 3.22$ (the 99% quantile): 23 days against 14.96 ($p_{\mathrm{MC}} = 0.030$)
  - the descriptive excess, 35.16 days, does not estimate the number of true drift bursts
  - the paper, on E-mini S&P 500: 202 drift bursts with $|T_t| > 4.5$ in 2010–2017, about one every two weeks
- 73 of the 109 days peak on the negative side (67%); the random-sign null is symmetric by construction
- The price reverses on 65% of the 80 days with a full window after the peak: a descriptive statistic
  - the peak is chosen in hindsight over the whole day, so the returns after it shape its choice
  - under the null, with the same selection: 56.5% on average; at least 65% in 12.4% of 1 000 replications
  - the size: -43.8 bp in the 15 minutes to the peak, then $+4.7$ bp
  - slope of R^+ on R^- (Eq. 36): 0.03 (t = 0.41); the paper, E-mini, $|T_t| > 4$: -0.26 (t = -12.05)
  - negative peaks, volume above the median (relative to its mean at the same time of day): -53.9 bp, then $+6.4$ bp (29 days)
  - negative peaks, volume below the median: -34.4 bp, then $+3.1$ bp (31 days)

Second Case Study (6/6): Validating the Procedure; Limits

- The same procedure on data simulated without drift bursts: 50 samples of 1 468 days, 5-minute bars
  - Heston with the VIX-based parameters, leverage ($\rho = -0.69$), one-minute steps
  - mean excess: 1.2 days (95% quantile: 14.8); $p_{\mathrm{MC}} \leq 0.05$ in 8% of the samples
  - plus asymmetric Merton jumps ($\hat{\lambda}$, $\hat{\mu}_J$, $\hat{\sigma}_J$ fitted to the S&P 500)
  - mean excess: 3.8 days (95% quantile: 17.5); $p_{\mathrm{MC}} \leq 0.05$ in 12% of the samples
  - mean reversal after the peak: 56% without jumps and 53% with jumps
- The excess in the data, 35.16 days, lies above the 95% quantile of both simulated models
- What the result supports: a description, compared with nulls without drift
  - it does not identify the paper's liquidity mechanism
- Limits
  - the power of the test at 5 minutes is not estimated here: low power remains an anticipated limitation
  - with 50 samples, the rejection frequencies carry a Monte Carlo error of about ± 4 percentage points
  - 29 days peak in the last 15 minutes of the session and have no window after the peak

An Open Question: How Much of the Price Path Is Discontinuous? (1/2)

- Model: $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t + J_t\,\mathrm{d}N_t$
  - X_t: the log price
  - N_t: the number of jumps (intensity λ)
  - J_t: the jump size
- The answer depends on the sampling frequency and on the model
  - Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance
  - The 5-minute bipower test (Chapter 9): jumps carry 2.8% of SPY variance

An Open Question: How Much of the Price Path Is Discontinuous? (2/2)

- Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear
  - they are rapid but continuous moves
- Christensen, Oomen & Renò (2022): **drift bursts** look like jumps in coarse data
  - drift burst: a short explosive trend in μ_t
- Formal question: H_0: $\lambda = 0$ against H_1: $\lambda > 0$
  - H_0: continuous path with stochastic volatility
  - a non-standard test, with the boundary problem of the Merton LR test (Section 9)

The Discovery Loop with an AI Assistant

The loop with an AI (Artificial Intelligence) assistant

Assistant: an LLM (Large Language Model) such as Claude, ChatGPT or Gemini.

- **1. Literature**: published jump shares by frequency
  - check every Digital Object Identifier (DOI) on Crossref
- **2. Hypotheses**: Merton jumps, Heston volatility, drift bursts
  - each with a distinct testable implication
- **3. Replicate first** the chapter's estimates
- **4. Robustness**: filters and tests fixed before the first run
- **5. Critique**: the assistant challenges the inference, as a referee would
- **6. Report**: every specification, the code, the AI log
- Reusable prompts: Appendix

What the Researcher Checks: Failure and Test

Checks: failure and test

- Boundary, unidentified parameters: parametric bootstrap under H_0
- Look-ahead in the volatility filter: standardise only with $\mathcal{F}_{t-1}$
- Frequency mismatch: jump shares at every frequency on a common sample
- Multimodal likelihood: many starting points and a profile in λ
- Invented numbers: every claim needs a table and a DOI

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- A typical assistant proposal: "71% of variance comes from jumps"
  - Rival hypothesis: the Poisson mixture absorbs time-varying volatility
  - then standardising by EWMA volatility removes the jumps
  - EWMA: Exponentially Weighted Moving Average, known at $t-1$
  - $\lambda_{\mathrm{EWMA}} = 0.94$, the RiskMetrics value for daily data

	Raw returns	EWMA-standardised
Excess kurtosis	11.0	2.9
LR, Merton vs the Normal distribution	2 403	619
$\hat{\lambda}$ per year (SE)	100 (10)	124 (53)
Jump share of variance	71%	51%

- The same sample: 8 995 days, without the first 250 (the EWMA start-up)
- Standardising cuts the excess kurtosis from 11.0 to 2.9, yet $\hat{\lambda}$ rises from 100 to 124 per year
  - EWMA need not remove heteroskedasticity, and does not separate jumps from stochastic volatility mechanically
- Interpretation: the estimated jump intensity and jump share depend strongly on how volatility is specified
- This exercise does not identify discontinuities structurally
- What the result supports: comparison of procedures
  - the same Merton model, fitted to raw and to EWMA-standardised returns

Your Turn: A Project Seed

- Questions
  - how much of S&P 500 and Bitcoin variance is discontinuous?
  - how does this share depend on the sampling frequency?
  - are the detected "jumps" drift bursts?
- Data: 5-minute bars for every intraday frequency
  - SPY: October 2020 – September 2026, 1 468 full trading days
  - Bitcoin: September 2024 – September 2026, 659 UTC days with at least 95% of the 288 bars (Chapter 9)
  - S&P 500 and Bitcoin daily closes
- The Bitcoin bars cover two years
  - an intraday comparison of the two assets uses the common window September 2024 – September 2026
- A pre-registration dated before the first run
  - contents: hypotheses, frequencies, filters, tests, bootstrap design

Key Takeaways and Next: Chapter 12 (1/2)

- Rescaled random walks converge to Brownian motion (Donsker)
  - its quadratic variation t drives Itô calculus
- Itô's lemma gives exact solutions (GBM, Ornstein–Uhlenbeck, Merton)
  - other models are simulated with Euler–Maruyama or Milstein at a known error order
- Frequent sampling pins down volatility
  - drift and mean-reversion speed need long spans
  - $\hat{\kappa}$ is biased upwards
- No arbitrage makes prices discounted Q-expectations
  - Girsanov changes only the drift
  - Feynman–Kac turns the expectation into a PDE

Key Takeaways and Next: Chapter 12 (2/2)

- $(\mathrm{VIX}/100)^2$ is a Q-expectation, affine in v_t
- In the fitted models jumps help with the tails, stochastic volatility with the clustering
- Volatility looks rough at short lags and persistent at long lags
  - a one-factor OU model does not reproduce this
- Next, Chapter 12: pricing options by replication
  - and reading the volatility surface

Check Your Understanding

- Why does a change of variable need an extra term in Itô calculus?
  - Identify the role of quadratic variation.
- Does observing prices more frequently make the drift estimate precise?
  - Separate frequency from the length of the observation period.
- Which feature calls for jumps, and which calls for stochastic volatility?
  - Use the comparison with observed returns.
- Why can a pricing probability differ from a real-world probability?
  - State what changes under the change of measure.

The Practical Decision

- **What I choose**
  - for daily S&P 500 returns, Bates (stochastic volatility with jumps), as the next model to fit and validate
  - the Heston part gives the clustering: ACF of $|r_t|$ at lag 1 is 0.28 against 0.27 in the data
  - the Merton part gives the tail: Hill statistic 2.70 against 2.56
  - $\hat{\lambda}$ describes fat tails, not the number of discontinuities
  - volatility from frequent sampling; drift and κ only from long spans, reported with their CIs
- **Under which conditions**
  - parameters under P for risk and under Q (VIX, option surfaces) for pricing
  - simulation with many steps per day (390 for Heston): one step removes the effect of ρ on daily skewness
- **What result would make me drop it**
  - a simulated Hill statistic or ACF of $|r_t|$ from the fitted model far from the values in the data
  - volatility autocorrelation at long lags above that of one factor, at the horizon that matters
  - in SPY: 0.13 at lag 100 against 0.002 for an OU fit; then I choose a rough, persistent model

Data Used in This Chapter (1/2)

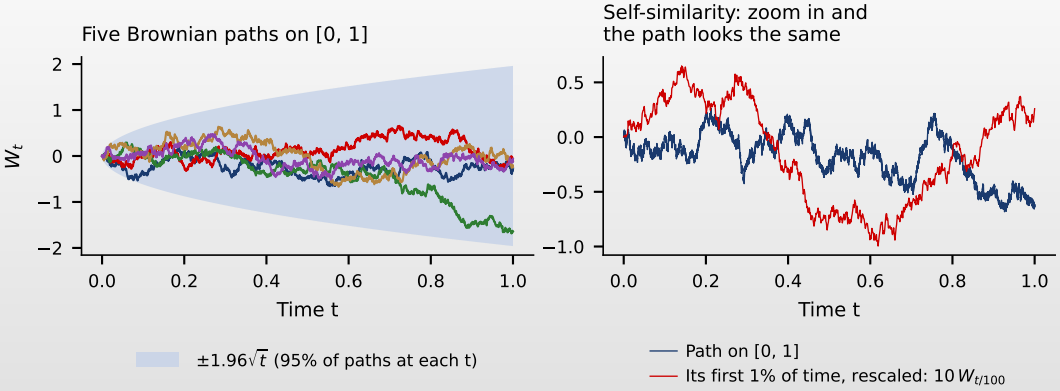
Series	Type	Period	Observations / frequency
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026
  - daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar
  - cleaning rules: Appendix
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: $1/365$)
  - drifts and volatilities annualised

Data Used in This Chapter (2/2)

- Intraday: SPY 5-minute bars, October 2020 – September 2026, 1 468 full trading days
  - SPY: exchange-traded fund (ETF) tracking the S&P 500
  - used in the case study; see Chapter 9
- Intraday: Bitcoin 5-minute bars, September 2024 – September 2026, 659 UTC days
  - days with at least 95% of the 288 bars, as in Chapter 9
- Notation
  - W_t is standard Brownian motion and S_t (also P_t) the price
  - v_t is the instantaneous variance
  - $\mathcal{F}_t$ is the information (filtration) up to t
  - $\mathrm{d}S/S$ is the instantaneous simple return and $\mathrm{d}\ln S$ the log change
  - Itô's lemma explains their different drifts

Brownian Paths (1/2)



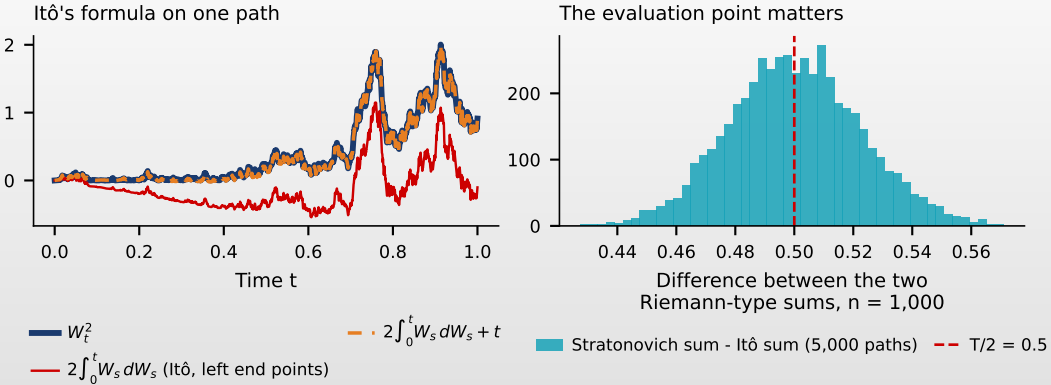
MFM_ch11_bm_paths

Brownian Paths (2/2)

- Left: five paths with the pointwise band $\pm 1.96\sqrt{t}$
  - the band covers 95% for each t separately, not for the whole path
- Right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of Chapter 7)
- A zoomed piece looks like the whole path
  - real markets break self-similarity
  - fat tails and clustering depend on the horizon (Chapter 1)

MFM_ch11_bm_paths

Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (1/2)



MFM_ch11_ito

Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (2/2)

- Result: $\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
  - each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2}(\Delta W_k)^2$
  - summed over k, the first parts give $\frac{1}{2}(W_T^2 - W_0^2) = \frac{1}{2} W_T^2$: the intermediate terms cancel
  - the second parts tend to $\frac{1}{2} T$
  - ordinary calculus would give $\frac{1}{2} W_T^2$: the extra $-\frac{1}{2} T$ is the quadratic variation
  - the mean is $\frac{1}{2}(T - T) = 0$
- Left: on one path of 2 000 steps, $2 \int_0^t W \, \mathrm{d}W + t$ lies approximately on W_t^2 (largest gap 0.038)
  - the exact identity on the grid: $W_t^2 = 2 \sum_i W_{t_i} \Delta W_i + \sum_i (\Delta W_i)^2$; t is only the limit of the discrete quadratic variation
- Right: Stratonovich minus Itô sums over 5,000 paths
  - mean 0.500 and standard deviation 0.022
  - theory gives $T/2 = 0.5$
  - Itô sums: mean 0.007 (theory: 0)
  - Itô sums: variance 0.520 (theory, by the isometry: $\int_0^1 t \, \mathrm{d}t = 1/2$)

MFM_ch11_ito

Quadratic Covariation and the Multivariate Itô Formula (1/2)

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \, \Delta Y_k$
  - if $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$, then $\mathrm{d}[W^{(1)}, W^{(2)}]_t = \rho \, \mathrm{d}t$
  - product rule: $\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + \mathrm{d}[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$
  - $f(t, x)$ continuously differentiable once in t and twice in x
  - $\mathrm{d}f(t, X_t) = f_t \, \mathrm{d}t + \sum_i f_{x_i} \, \mathrm{d}X_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} \, \mathrm{d}[X^i, X^j]_t$
  - X_t^i is the i-th component

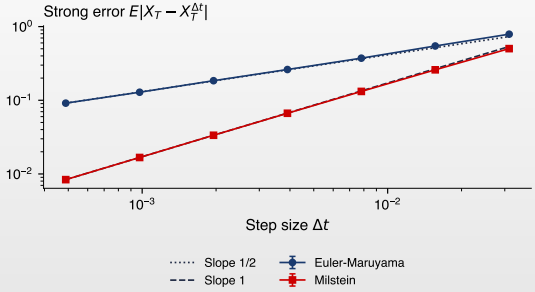
Quadratic Covariation and the Multivariate Itô Formula (2/2)

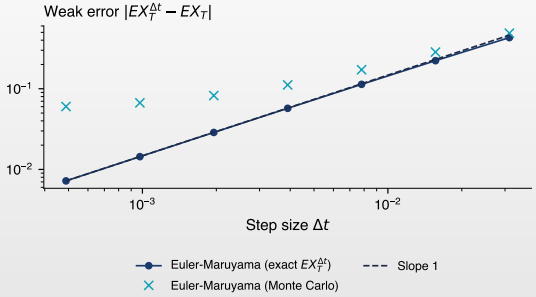
- Example (Heston, Section 10)
  - $\mathrm{d}S = \mu S\,\mathrm{d}t + \sqrt{v}\,S\,\mathrm{d}W^{(1)}$
  - $\mathrm{d}v = \kappa(\theta - v)\,\mathrm{d}t + \xi\sqrt{v}\,\mathrm{d}W^{(2)}$
  - v is the instantaneous rate of quadratic variation of $\ln S$ (the instantaneous variance of returns), not $\mathrm{Var}(S_t)$
  - κ is the speed of mean reversion
  - θ is the long-run variance and ξ the volatility of the variance
- $\mathrm{d}(S_t v_t) = S_t\,\mathrm{d}v_t + v_t\,\mathrm{d}S_t + \rho\,\xi\,v_t S_t\,\mathrm{d}t$
  - ρ enters the drift of the product

Weak Convergence

- **Weak error**: $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C\,\Delta t^{\beta}$
  - how close the simulated distribution is to the true one
  - g is a function of the final value (e.g. an option pay-off)
  - β is the weak order
- Both schemes have $\beta = 1$
  - enough for European option prices and other expected values
  - conditions: smooth coefficients with bounded derivatives and a smooth g Kloeden & Platen (1992)
- Slower convergence is possible
  - square-root coefficients (CIR (Cox–Ingersoll–Ross), Heston)
  - quantiles such as Value-at-Risk (VaR)
  - barrier crossings
- Test equation Higham (2001): $\mathrm{d}X = \lambda X\,\mathrm{d}t + \mu X\,\mathrm{d}W$
  - $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark
  - on a log-log plot, the order is the slope of the error against the step (next slide)

Convergence of the Schemes (1/2)





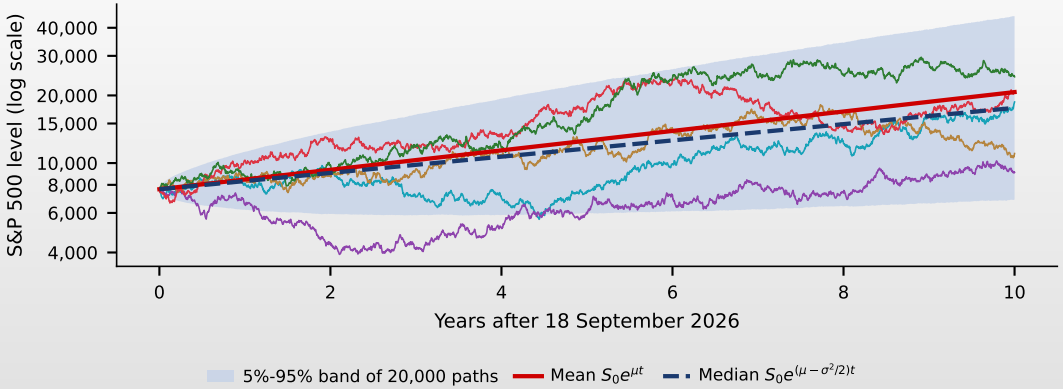
MFM_ch11_convergence

Convergence of the Schemes (2/2)

- 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5}
- Strong orders: slope of the log-log fit, 95% CI by resampling whole paths
  - Euler–Maruyama 0.518 [0.512, 0.525], Milstein 0.986 [0.984, 0.987]
  - the narrow CIs measure Monte Carlo noise only
  - the small gap to 1/2 and 1 is finite-step curvature, not a different order
  - halving Δt halves the Milstein error, but cuts the Euler error only by 29%
- Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985
  - a deterministic slope
  - the Monte Carlo estimates flatten at small steps
  - their standard error, about 0.068, exceeds the bias
  - the sample size, not the step, is often the binding constraint Glasserman (2003)

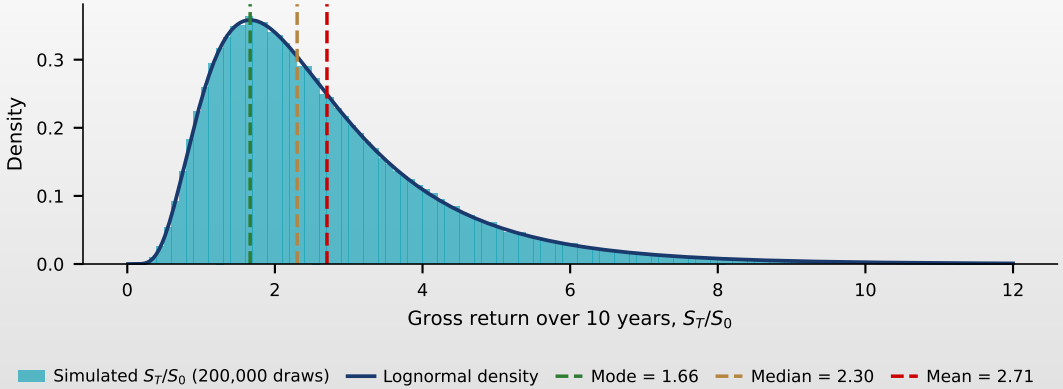
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Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (1/3)



MFM_ch11_gbm_fan

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (2/3)



MFM_ch11_gbm_fan

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (3/3)

- GBM with $\hat{\mu} = 10.0\%$ and $\hat{\sigma} = 18.0\%$ (1990–2026), over $T = 10$ years
  - mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, mode $S_0 e^{(m-\sigma^2)T}$
  - density (200 000 draws): the gross return S_T/S_0 is 2.71 at the mean
  - it is 2.30 at the median and 1.66 at the mode
- Fan chart: 20,000 exact paths starting from $S_0 = 7\,650$ (18 September 2026)
  - 90% of the paths end between 6 878 (5% quantile) and 45 017 (95% quantile)
  - probability of a 10-year loss: $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
- $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$
  - most investors end below the average
  - so report the median and quantiles
- The fan is too narrow in crises
  - GBM has no fat tails and no volatility clustering (tested on the S&P 500)

MFM_ch11_gbm_fan MFM_ch11_lognormal

Recap: Growth, Lognormality and Drift Precision

- Formulas to keep
  - $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with $m = \mu - \frac{1}{2}\sigma^2$
  - the median grows at m, the mean $\mathbb{E}[S_T]$ at μ
  - $\mathrm{SE}(\hat{\sigma}) = \sigma/\sqrt{2n}$ shrinks with the number of returns
  - $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$ shrinks only with the years covered
- Check: with $\sigma = 20\%$, what is the SE of the log drift after 10 years of data?
- Check: and after 40 years?
  - Answer: $0.20/\sqrt{10} = 6.3$ and $0.20/\sqrt{40} = 3.2$ percentage points
  - four times the span only halves the error, whatever the sampling frequency

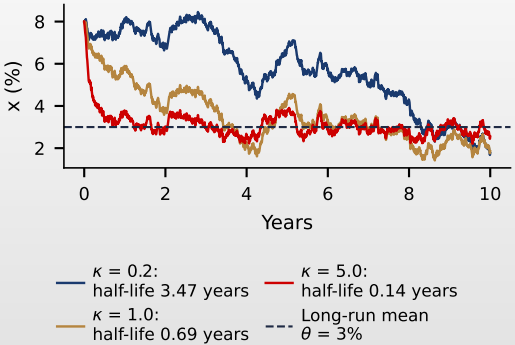
Solving the OU Equation in Three Steps (1/2)

- Step 1: Itô's lemma on $f(t,x) = e^{\kappa t}x$ ($f_{xx} = 0$)
  - result: $\mathrm{d}(e^{\kappa t}x_t) = \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t$
  - the terms in x_t cancel
- Step 2: integrate from 0 to t and multiply by $e^{-\kappa t}$
  - $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s$
- Step 3: the Itô integral has mean 0
  - by the isometry, its variance is $\sigma^2 \int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s$
  - so x_t is Normal

Solving the OU Equation in Three Steps (2/2)

- Moments of the Normal x_t
  - $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$
  - $\text{Var}(x_t \mid x_0) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right)$
- Stationary distribution $N\left(\theta, \sigma^2/(2\kappa)\right)$
  - unlike Brownian motion, the variance stops growing with the horizon
- Check: what are the mean and the standard deviation of x_1?
  - inputs: $x_0 = 8\%$, $\theta = 3\%$, $\kappa = 0.5$, $\sigma = 1\%$
  - Answer, mean: $3\% + 5\% \cdot e^{-0.5} = 6.03\%$
  - standard deviation: $1\% \cdot \sqrt{1 - e^{-1}} = 0.80\%$
  - after one year 61% of the initial gap remains

Three Speeds of Mean Reversion



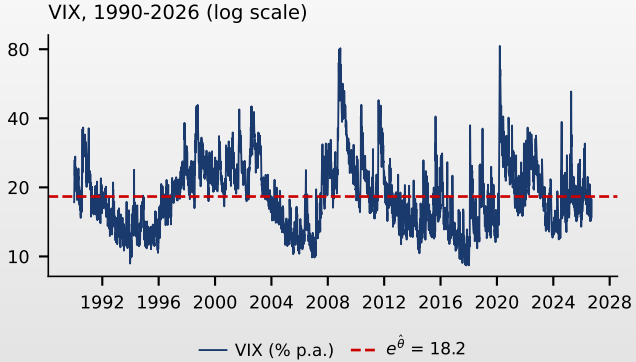
- OU paths driven by the same shocks, starting at 8%
  - $\theta = 3\%$, $\sigma = 1\%$, and speed $\kappa = 0.2$ (slow), 1 or 5 (fast)
  - exact simulation, daily steps
- Half-lives: 3.47 years for $\kappa = 0.2$, 0.69 for $\kappa = 1$ and 0.14 for $\kappa = 5$
  - With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$
  - 1.58% for $\kappa = 0.2$, 0.71% for $\kappa = 1$ and 0.32% for $\kappa = 5$
  - Slow reversion looks like a random walk for years
- Long-horizon forecasts converge to θ

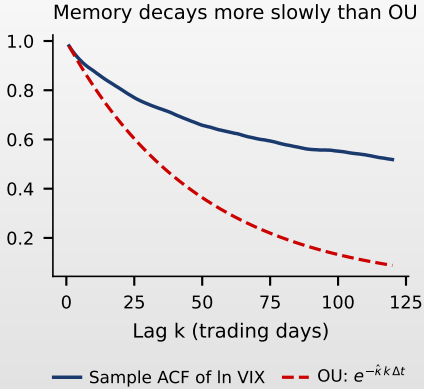
MFM_ch11_ou_paths

Why Estimated Mean Reversion Looks Too Fast

- In finite samples $\hat{\kappa}$ overstates κ by roughly $4/T$ (T in years) Tang & Chen (2009)
  - Simulation: the full-sample estimates as the truth, 500 samples of 72 years
  - The mean $\hat{\kappa}$ is 0.164, a bias of 0.063, close to the approximation $4/T = 0.055$
  - 78% of the estimates exceed the truth (Seminar 11, B2)
- Where the bias comes from: $\hat{b}$ is biased towards zero in a short persistent sample
  - $\hat{b}$: the OLS slope of $x_{t+\Delta t}$ on x_t
  - $\hat{\kappa} = -\ln \hat{b}/\Delta t$ turns a too-small $\hat{b}$ into a too-large $\hat{\kappa}$
- The bias depends on the span T, not on the number of observations
  - sampling daily instead of monthly does not remove it (Section 7)

Is Volatility Mean-Reverting? The VIX (1/2)





MFM_ch11_vix_ou

Is Volatility Mean-Reverting? The VIX (2/2)

- Left: VIX 1990–2026 on a log scale with the OU long-run level $e^{\hat{\theta}} = 18.2$
  - OU fit to $\ln \mathrm{VIX}$: $\hat{\kappa} = 5.10$ (SE 0.53), a half-life of 34 trading days
  - Spikes such as 82.7 (2020-03-16) fade within weeks
- Right: sample ACF of $\ln \mathrm{VIX}$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$
  - At lag 60, the autocorrelation is 0.63 in the data but 0.30 under OU
  - after the fast decay a slow component remains
- One OU factor is too simple
  - candidates: two-factor or long-memory models
  - rough volatility (Section 12) is a distinct alternative
- Mean-reverting volatility is the building block of Heston

MFM_ch11_vix_ou

Recap: Mean Reversion in Three Formulas

- Formulas to keep
  - Expected path: $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$
  - Half-life: $\ln 2/\kappa$
  - Long-run spread: stationary standard deviation $\sigma/\sqrt{2\kappa}$
  - Estimation: AR(1) slope $b = e^{-\kappa \Delta t}$, hence $\hat{\kappa} = -\ln \hat{b}/\Delta t$
  - $\hat{\kappa}$ is biased upwards in short samples
- Check: monthly data give $\hat{b} = 0.95$. What are $\hat{\kappa}$ and the half-life?
  - Answer: $\hat{\kappa} = -12 \ln 0.95 = 0.616$ per year
  - half-life: $\ln 2/0.616 = 1.13$ years, about 13.5 months

Likelihood for Nonlinear Diffusions (1/2)

- **Euler pseudo-likelihood**: a Gaussian transition
  - $r_{t+\Delta} \mid r_t \sim N\left(r_t + a(r_t)\Delta,\ b^2(r_t)\Delta\right)$
  - $a(r)$, $b(r)$: the drift and diffusion of the rate r
  - its bias, of order Δ, is small with daily data
- **Exact CIR**: $2c\, r_{t+\Delta} \mid r_t$ is noncentral χ^2 Cox, Ingersoll & Ross (1985)
  - $4\kappa\theta/\sigma^2$ degrees of freedom, $c = 2\kappa/\left(\sigma^2(1 - e^{-\kappa\Delta})\right)$
  - The noncentrality parameter is $2c\, r_t e^{-\kappa\Delta}$
  - The Feller condition is equivalent to at least 2 degrees of freedom (Seminar 11, A7)

Likelihood for Nonlinear Diffusions (2/2)

- General diffusions: closed-form Hermite expansions of the transition density Aït-Sahalia (2002)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma r^{\gamma}\,\mathrm{d}W$ Chan, Karolyi, Longstaff & Sanders (1992)
  - nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$)
  - estimated by Gaussian (Euler) quasi-likelihood
  - Tests on γ use sandwich standard errors Newey & West (1987)
  - the score covariance is heteroskedasticity- and autocorrelation-consistent (HAC)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007

Model	$\hat{\kappa}$ daily (SE)	$\hat{\kappa}$ monthly (SE)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033) daily; 0.60 (0.149) monthly

- Data: daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$)
- 2008–2026 is excluded (Appendix)
  - exact zeros and negative quotes lie outside the domain of these likelihoods
- Exact and Euler differ by less than one standard error
  - discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- Volatility rises with the level (statistics: Appendix)
  - HAC Wald tests reject $\gamma = 0$ at both frequencies
  - they reject $\gamma = 1/2$ with daily but not with monthly data

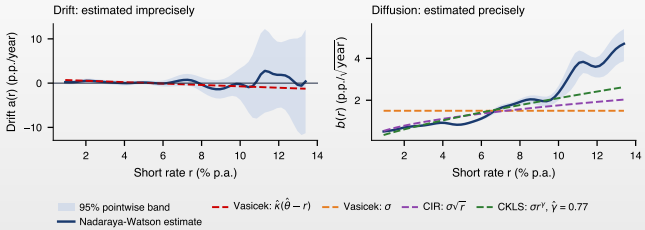
Nonparametric Drift and Diffusion (1/2)

- No functional form: kernel averages of the increments around each level x Stanton (1997)
  - $\Delta r_t = r_{t+\Delta} - r_t$ is the next increment
  - $w_t(x)$ is the weight of day t, large when r_t is close to x
  - $\hat{a}(x) = \sum_t w_t(x)\,\Delta r_t/\Delta$
  - $\hat{b}^2(x) = \sum_t w_t(x)\,(\Delta r_t)^2/\Delta$
  - Nadaraya–Watson weights $w_t(x) = K\left((r_t - x)/h\right) / \sum_s K\left((r_s - x)/h\right)$
  - K: Gaussian kernel; h: bandwidth

Nonparametric Drift and Diffusion (2/2)

- The moment conditions are exact only as $\Delta \to 0$
  - $\mathbb{E}[\Delta r \mid r = x]/\Delta = a(x) + O(\Delta)$
  - $\mathbb{E}[(\Delta r)^2 \mid r = x]/\Delta = b^2(x) + O(\Delta)$
- The diffusion converges at the infill rate, the drift only at the long-span rate
- The results hold for non-stationary but recurrent processes Bandi & Phillips (2003)
- Caution: nonparametric specification tests reject far too often with persistent data Pritsker (1998)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Data: daily DTB3, 1954–2007, bandwidth $h = 0.44$ percentage points
  - Silverman's rule, $h = 1.06\,\hat{s}\,n^{-1/5}$, with $\hat{s}$ the standard deviation of the rate
- The 95% pointwise bands are heteroskedasticity-robust but ignore serial dependence
- Drift: zero lies inside the band at 97% of the levels
- Diffusion: the median relative SE is 8%
  - $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits at every level
  - share of levels inside the band: CKLS 35%, CIR 7%, Vasicek 5% (descriptive shares)

MFM_ch11_np_diffusion

Recap: What Frequency Buys and What It Cannot

- Formulas to keep
  - Infill: $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$
  - so the diffusion coefficient is learned from frequent sampling
  - Long span: $I(m) = T/\sigma^2$ and $\mathrm{Var}(\hat{\kappa}) \approx 2\kappa/T$
  - so drift and speed are learned only from more years
  - Bias: $\hat{\kappa}$ exceeds κ by roughly $4/T$
- Check: which standard error shrinks, that of $\hat{\sigma}$ or that of $\hat{\kappa}$?
  - setting: the same 20 years sampled daily instead of monthly (21 times more observations)
  - Answer: that of $\hat{\sigma}$, by about $\sqrt{21} \approx 4.6$ times
  - that of $\hat{\kappa}$ barely moves, since $2\kappa/T$ does not depend on the step

Moments of the Merton Return (1/2)

- Cumulants c_k of the return over Δt: $K(u) = \ln \mathbb{E}[e^{ur}]$ and $c_k = K^{(k)}(0)$
  - c_1 mean, c_2 variance, c_3 third central moment
  - cumulants of independent parts add
  - a Normal part has $c_k = 0$ for $k \geq 3$
  - The k-th cumulant of a compound Poisson sum is $\lambda \Delta t \, \mathbb{E}[Y^k]$
- Merton cumulants
  - $c_1 = m\Delta t + \lambda \Delta t \, \mu_J$
  - $c_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$
  - $c_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
  - $c_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$

Moments of the Merton Return (2/2)

- Skewness $c_3/c_2^{3/2}$ is negative when $\mu_J < 0$
- Excess kurtosis c_4/c_2^2 is positive with any jump
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows
  - jumps matter most at short horizons
- Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$ and $\sigma_J = 4\%$ (Seminar 11, A3)
  - the daily excess kurtosis is 17.6
  - the skewness is -2.07

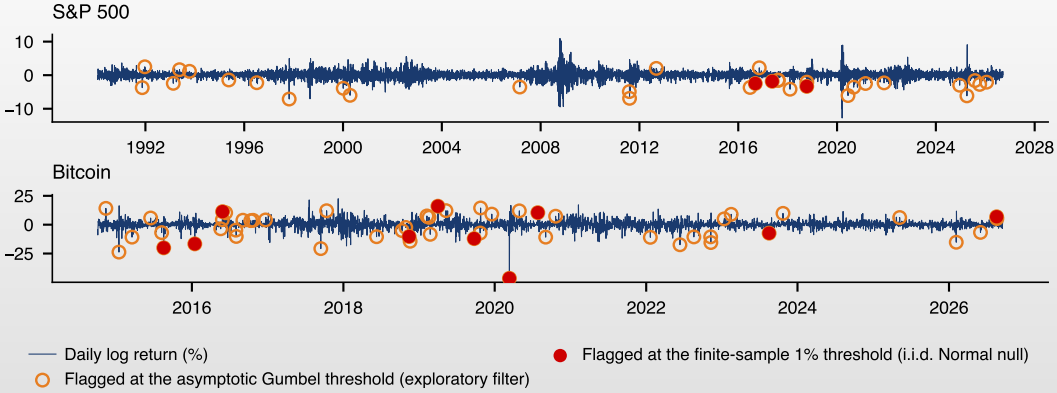
Detecting Jumps: the Lee–Mykland Test (1/2)

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
  - $L_t = r_t/\hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2}\sum_{j=t-K+2}^{t-1} |r_j||r_{j-1}|$
  - $\hat{\sigma}_t^2$ is the bipower variation, robust to jumps
  - With daily data the window is $K = 16$ days, as recommended by the authors
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
  - Reject if $(|L_t| - C_n)/S_n > -\ln(-\ln(1-\alpha))$
  - the threshold is 4.60 at $\alpha = 1\%$ for the **normalised** statistic, not for $|L_t|$

Detecting Jumps: the Lee–Mykland Test (2/2)

- Normalising constants
  - $C_n = \frac{\sqrt{2 \ln n}}{c} - \frac{\ln \pi + \ln \ln n}{2c\sqrt{2 \ln n}}$, $S_n = \frac{1}{c\sqrt{2 \ln n}}$
  - $c = \sqrt{2/\pi}$: the mean of $|Z|$, which rescales the bipower estimate
  - $n =$ number of tested days
- The test controls, asymptotically, the probability of a false jump over the **whole** sample
  - not day by day

Days Flagged by the Lee–Mykland Test in the S&P 500 and Bitcoin (1/2)



MFM_ch11_jumps

Days Flagged by the Lee–Mykland Test in the S&P 500 and Bitcoin (2/2)

	S&P 500	Bitcoin
Flagged, asymptotic threshold	32	52
Per year	0.87	4.35
Negative (%)	84	52
In the top 10 moves	0	5
Above the calibrated 1% threshold	3	10

- Daily log returns; the asymptotic 1% threshold works here as an exploratory filter
- S&P 500: the largest moves (2008, 2020) are not flagged
  - they came in turbulent periods
  - Instead, a 1.2% move in a calm market is flagged (1993-10-14)
- The flagged days differ from the fitted Merton intensity. Possible explanations:
  - stochastic volatility
  - drift bursts (short explosive drift episodes)
  - low detection power in daily data

MFM_ch11_jumps

The Lee–Mykland Threshold in Finite Samples

- The asymptotic threshold assumes a precisely estimated local volatility; with $K = 16$ days, $\hat{\sigma}_t$ is noisy
  - in 2 000 i.i.d. samples from the Normal distribution, of the S&P 500 length, the threshold 4.60 flags at least one day in 96.6% of the samples
  - on average 3.43 flagged days per sample, although there are no jumps
- Calibrated threshold: the 99% quantile of the sample maximum of the statistic in the same samples (same n, same K)
  - S&P 500: 22.93 ($|L_t| > 11.6$), against 4.60 ($|L_t| > 6.2$); 3 flagged days (2016-09-09, 2017-05-17, 2018-10-10)
  - Bitcoin: 19.61 ($|L_t| > 10.6$); 10 flagged days
- Size under stochastic volatility without jumps: 200 Heston samples (VIX-based parameters, $\rho = -0.69$, 390 steps per day)
  - the asymptotic threshold is exceeded in 100% of the samples
  - the threshold calibrated under i.i.d. Normal returns is exceeded in 44% of the samples
  - with the 99% quantile of the Heston samples (48.7), no S&P 500 day is flagged
- With daily data, the flagged days are candidates for inspection, not jumps detected at the 1% level

What the VIX Identifies in Heston

- Under Q: $y_t = \frac{1}{\tau}\mathbb{E}^Q\left[\int_t^{t+\tau} v_s \, \mathrm{d}s \mid \mathcal{F}_t\right] = a + b\, v_t$
  - $\tau = 30/365$ (Seminar 11, A8)
  - $b = \frac{1-e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1-b)$
  - y_t is **affine** in v_t, with Q-parameters
- What the regression on y_t recovers
  - κ: the P-speed, since an affine map keeps the dynamics of v_t
  - ρ: unchanged (correlation is scale-free), but only approximated by daily proxy changes
  - ξ is attenuated (derivation: next slide)
  - the long-run level mixes θ^Q and θ^P
- Proper estimation: one of two routes
  - fit the dynamics of y directly, with diffusion proportional to $\sqrt{y-a}$
  - or fit v_t and the observation equation jointly

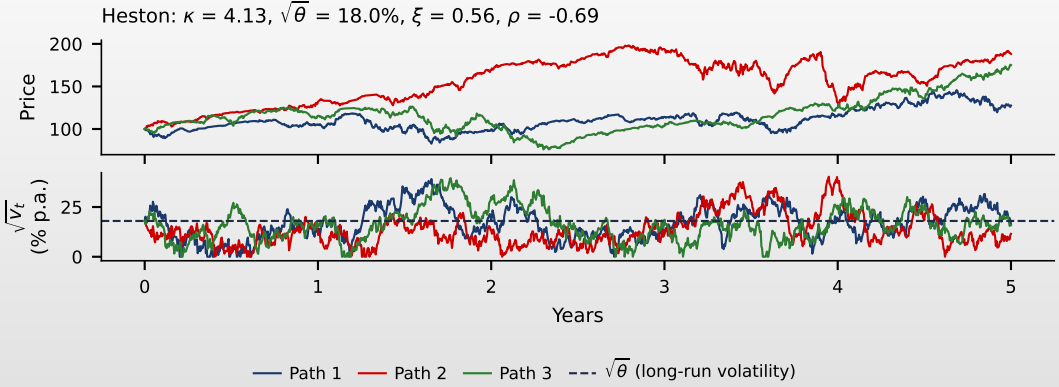
Where the Attenuation of ξ Comes From (1/2)

- Step 1: $y_t = a + b\,v_t$ with constants a, b, so $\mathrm{d}y_t = b\,\mathrm{d}v_t$ under P
  - $\mathrm{d}y_t = b\kappa(\theta^P - v_t)\,\mathrm{d}t + b\,\xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$
- Step 2, drift: $b\kappa(\theta^P - v_t) = \kappa(\theta_y - y_t)$ with $\theta_y = a + b\,\theta^P$
  - the speed κ is unchanged
  - $\theta^P = (\theta_y - a)/b$
- Step 3, diffusion: the CIR regression assumes $\xi_y\sqrt{y_t}$
  - matching $\xi_y\sqrt{y_t} = b\,\xi\sqrt{v_t}$ gives $\xi_y = b\,\xi\sqrt{v_t/(a + b\,v_t)}$
  - at $v_t = \theta^Q$, $a + b\,\theta^Q = \theta^Q$, so $\xi_y = b\,\xi$

Where the Attenuation of ξ Comes From (2/2)

- Numbers: with $\kappa^Q = 5$, $b = 0.82$
  - the fitted $\hat{\xi} = 0.56$ corresponds to $\hat{\xi}/b = 0.68$ near $v_t = \theta^Q$
  - other κ^Q: Appendix
  - the correction $\hat{\xi}/b$: Appendix
- κ^Q is not identified from the VIX alone
  - joint P/Q estimation with option prices Chernov & Ghysels (2000)
  - or moments of realised variance Bollerslev & Zhou (2002)

Simulated Heston Paths (1/2)



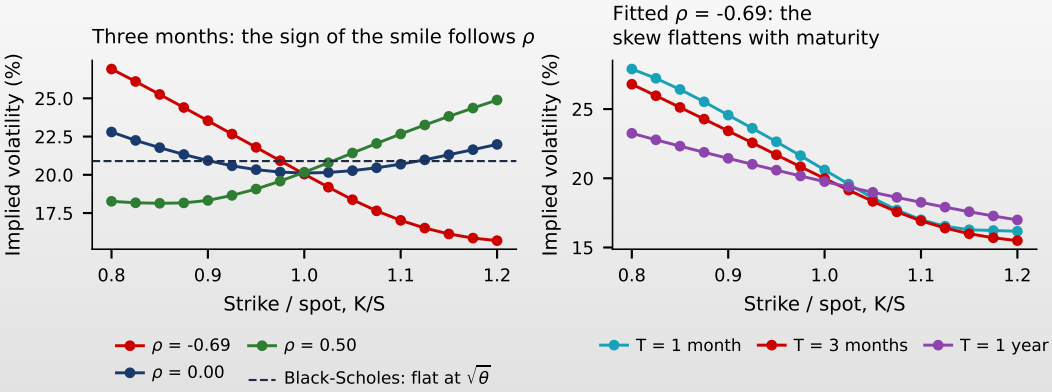
MFM_ch11_heston_paths

Simulated Heston Paths (2/2)

- Three five-year paths: top, the price; bottom, the volatility $\sqrt{v_t}$
  - $\hat{\kappa}$, $\hat{\xi}$ (uncorrected) and $\hat{\rho}$ come from the VIX
  - θ is the realised S&P 500 variance
- Volatility clustering now comes from the model
  - Price falls coincide with volatility spikes
  - Simulated return–variance correlation: -0.69 (leverage through ρ)
- Feller ratio $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the variance can reach zero
  - the latent Euler state $\tilde{v}$ is ≤ 0 on 1.4% of the steps
  - full truncation: $\tilde{v}_{k+1} = \tilde{v}_k + \kappa(\theta - \tilde{v}_k^+)\Delta + \xi\sqrt{\tilde{v}_k^+\Delta}\, Z_k$, with $\tilde{v}^+ = \max(\tilde{v}, 0)$
  - $\tilde{v}^+$ enters the drift, the diffusion and the price step; $\tilde{v}^+$ is reported

MFM_ch11_heston_paths

A Preview of Chapter 12: the Volatility Smile (1/2)



MFM_ch11_heston_smile

A Preview of Chapter 12: the Volatility Smile (2/2)

- **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price
  - flat across strikes under GBM
  - here from Heston call prices (Monte Carlo, 100,000 antithetic paths)
- Left, three months to maturity
  - $\hat{\rho} = -0.69$, a downward **skew**: 26.9% at a strike of 80% of spot, 20.0% at the money, 15.7% at 120%
  - $\rho = 0$, a symmetric **smile**: 22.8% at 80% of spot, 20.1% at the money, 22.0% at 120%
  - $\rho = 0.50$, the skew reverses: implied volatility rises with the strike, from 18.3% to 24.9%
- Right: the skew flattens with maturity
  - one month: 27.9% – 16.2%
  - one year: 23.2% – 17.0%
- Illustration only: pricing needs κ^Q, θ^Q calibrated to option prices (Chapter 12)

MFM_ch11_heston_smile

Affine Jump-Diffusions

Affine jump-diffusions Duffie, Pan & Singleton (2000)

- Drift, variance, jump intensity affine in the state X_t
  - Vasicek, CIR, Merton, Heston and Bates
- $\mathbb{E}\left[e^{iu^\top X_T} \mid X_t\right] = \exp\left(\alpha(\tau) + \beta(\tau)^\top X_t\right)$
  - $i = \sqrt{-1}$, u: a real vector, $\tau = T - t$
  - α, β: solutions of Riccati ODEs
- Option prices by Fourier inversion
  - Simplest case: the Vasicek $A(\tau)$, $B(\tau)$ of Section 8

From GARCH to Diffusions

GARCH (Generalised Autoregressive Conditional Heteroskedasticity) models conditional variance in discrete data.

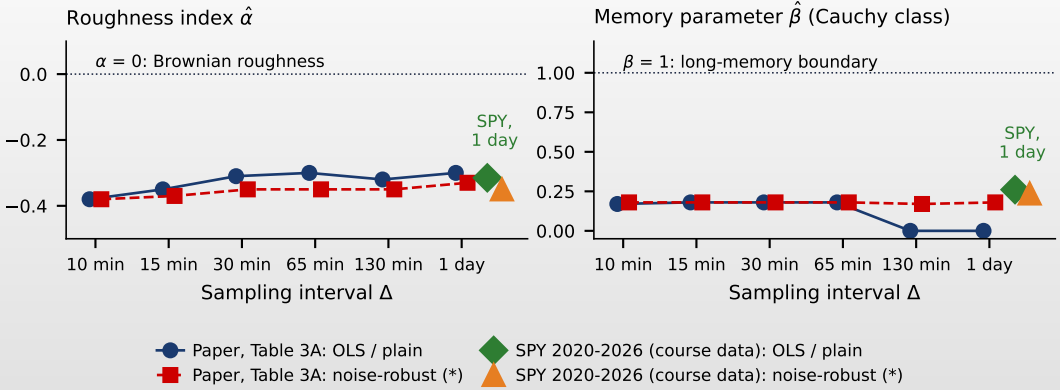
From GARCH to continuous time Nelson (1990)

- As the step shrinks, GARCH(1,1) converges to
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi v_t\,\mathrm{d}W_t^{(2)}$
  - Like Heston, with v_t instead of $\sqrt{v_t}$
- $\kappa \approx (1-\alpha-\beta)/\Delta t$
  - α, β: GARCH(1,1) coefficients
  - daily $\alpha+\beta = 0.99$: $\kappa \approx 2.5$ per year (Chapter 5)
  - half-life: about 70 days
- GARCH and its diffusion limit are not asymptotically equivalent statistical experiments Wang (2002)
  - inference can differ even as $\Delta t \to 0$

Why a Negative ρ Skews Daily Returns

- Over one day $h = 1/252$: $R = \int_0^h \sqrt{v_s}\,\mathrm{d}W_s^{(1)}$
  - drift terms of order h are ignored
  - Itô's lemma on $\sqrt{v}$: $\mathrm{d}\sqrt{v_s} = \frac{1}{2}\xi\,\mathrm{d}W_s^{(2)} + O(\mathrm{d}s)$
  - so $\sqrt{v_s} \approx \sqrt{v_0} + \frac{1}{2}\xi W_s^{(2)}$
  - $R \approx \sqrt{v_0}\,W_h^{(1)} + \frac{1}{2}\xi \int_0^h W_s^{(2)}\,\mathrm{d}W_s^{(1)}$
- Third moment: the cross term gives $\mathbb{E}[R^3] \approx \frac{3}{2}\,\xi\rho\,v_0 h^2$
  - while $\mathrm{Var}(R) \approx v_0 h$
  - skewness $\approx \frac{3}{2}\,\rho\,\xi\sqrt{h}/\sqrt{v_0}$, a leading-order approximation
  - with $\hat{\rho} = -0.69$, $\hat{\xi} = 0.56$ and $\sqrt{v_0} = 18.0\%$: about -0.20
  - the simulation with 390 steps per day (Section 11): median -0.21

Case Study: SPY Against Table 3 of the Paper (1/2)



MFM_ch11_rough_vol

Case Study: SPY Against Table 3 of the Paper (2/2)

- Lines: E-mini S&P 500 futures, 2011–2014 (Bennedsen, Lunde & Pakkanen (2022), Table 3, Panel A)
  - Δ from 10 minutes to 1 day
- Markers: SPY at 1 day, 2020–2026, course data
  - * marks the noise-robust estimators
- $\hat{\alpha}$ barely moves with Δ (-0.38 to -0.30)
  - roughness is a property of the process
  - SPY falls in the same band
- $\hat{\beta}^* = 0.17$–0.18 at every Δ, far below 1
  - the plain $\hat{\beta}$ is unstable
- Limits: another contract, period and estimator
  - bars, not every trade
  - no pre-averaging
- Realised measures estimate integrated, not spot, variance

MFM_ch11_rough_vol

Your Turn: Replication and Core Deliverables

- Replicate first
  - the Merton jump share of variance: 71% (daily S&P 500)
  - the days flagged by the Lee–Mykland test at the asymptotic threshold: 0.87 per year
  - the bipower jump share: 2.8% (SPY, 5 minutes, Chapter 9)
- Core deliverables, feasible with the course data
  - SPY: bipower jump share and Lee–Mykland flags, with thresholds calibrated by simulation, at 5, 15 and 30 minutes
  - The EWMA standardisation of the mini-case, repeated on SPY at each frequency

Your Turn: Drift Bursts in Bitcoin and Report

- Do drift bursts occur in Bitcoin, which trades 24/7, more or less often than in SPY?
  - second question: are they followed by a price reversal?
  - Data: Bitcoin 5-minute bars, September 2024 – September 2026, 659 UTC days with at least 95% of the 288 bars (Chapter 9)
  - the SPY 5-minute bars on the same window
  - before the first run: pre-register the bandwidths, the null (sign-randomised returns) and the comparison
  - explain how the absence of a closing auction and of an overnight gap changes the test
- Optional extension: a bootstrap LR test of Heston (null, no jumps) against Bates
  - both likelihoods need filtering of the latent variance
  - e.g. the Markov Chain Monte Carlo (MCMC) approach of Eraker, Johannes & Polson (2003)
  - plan the computing time before committing to it
- Deliverables
  - code and a two-page report with every specification tried
  - a referee-style critique of your own result
- Declare your AI use in `AI_USE.md` and the errors you caught in `AI_ERRORS.md`
  - github.com/danpele/MFM/tree/main/project

From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag
  - the observation is made under the microscope
- Bachelier (1900): a random walk in continuous time for prices
  - the prices: French government bonds (rentes) and their options
  - the spread of the price grows with $\sqrt{t}$
  - the result comes five years before physics
- Einstein (1905): molecular collisions explain Brown's motion
  - the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Back

Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC BY-SA 2.0 de; Wikimedia Commons



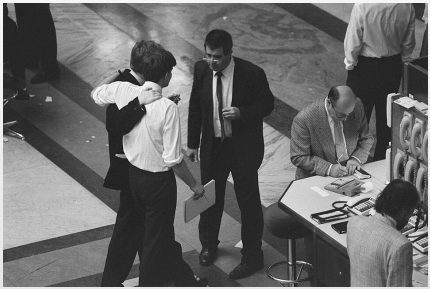
Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule
  - the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion
- Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility
  - textbook treatment: Franke, Härdle & Hafner (2019), Part I

Back

A Crash in Pictures, Amsterdam 1987



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- The fall of 19 October 1987 (Section 9): 22.6% in one day
- Under a Normal distribution such a day would not be expected once in the age of the universe
  - prices gap
  - continuous models need jumps or fat-tailed innovations

Back

Data Cleaning and Technical Conditions

- Data: equity indices on weekdays, stale quotes removed
  - stale quote: unchanged close and no intraday range
  - genuine zero returns are kept
  - Bitcoin: 7 days a week
- Itô integral: left-point sums converge for regular integrands
  - e.g. continuous and adapted integrands
  - a general H is reached by approximation with simple processes
  - if only $\int_0^T H_t^2 \, \mathrm{d}t < \infty$ almost surely, $\int H \, \mathrm{d}W$ is merely a local martingale
  - the mean of this local martingale need not be zero
- SDE: unique strong solution if a, b are Lipschitz in x with at most linear growth Kloeden & Platen (1992)
  - square-root coefficients (CIR, Heston) are not Lipschitz at zero
  - they still have unique strong solutions (Yamada–Watanabe conditions)
- Donsker: convergence in distribution in the space of càdlàg paths Billingsley (1999)
  - the space carries the Skorokhod topology

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Short Rates, More Numbers (1/2)

- DTB3, 1954–2026: the rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11)
  - 4.3% of the days were below 0.05%
  - The fitted Vasicek model gives a 6.1% long-run probability of a negative rate
  - 2008–2026: 13 exact zeros and 8 negative quotes
  - these quotes are outside the domain of the CIR and CKLS likelihoods
  - in this period, 17% of the days form a near-zero regime below 0.05%

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Short Rates, More Numbers (2/2)

- CKLS tests on γ, 1954–2007, HAC Wald with 11 (daily) and 6 (monthly) lags
  - $\gamma = 0$ rejected: daily 545, monthly 16
  - $\gamma = 1/2$ rejected daily: 67.0; 22.1 with 55 lags
  - $\gamma = 1/2$ not rejected monthly: 0.5 against 3.84
  - The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1)
  - the LR test ignores volatility clustering
  - Chan, Karolyi, Longstaff & Sanders (1992) found $\hat{\gamma} \approx 1.5$ for 1964–1989

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Vasicek Yield and the Observed Rate (1/2)

- Vasicek bond: 10-year yield 4.05%
  - with the Section 6 estimates, $\lambda = 0$ and $r = 3.99\%$
- The 10-year Treasury yield (FRED series DGS10) on 2026-09-18: 5.01%
  - DGS10 is a semiannual par yield
  - continuously compounded: $2\ln(1 + 5.01\%/2) = 4.95\%$
  - DTB3 is a 91-day discount rate: 3.99% is 4.07% continuously compounded
  - at this rate the model yield rises to 4.10%

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Vasicek Yield and the Observed Rate (2/2)

- The gap shrinks only from 0.96 to 0.85 percentage points
  - par versus zero-coupon yield is not adjusted
- The remaining gap cannot be attributed to a term premium
  - only the starting rate was converted, not the parameters estimated on the series
  - DTB3 is a 3-month proxy for the instantaneous rate
  - a structural comparison: convert the series before estimation and compare zero-coupon yields at the same maturity
- In the model, a positive term premium needs $\theta^Q > \hat{\theta}$, i.e. $\lambda < 0$: an implication of the model, not a result identified here
- Nonparametric estimation: Stanton (1997) also estimates the market price of interest-rate risk
- Specification test: Aït-Sahalia (1996)

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The VIX Correction by κ^Q and the Full Model Comparison

κ^Q	1	2.5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.21
Skewness	−0.36	0.00	−0.32	−0.21
Hill statistic	2.56	4.64	2.70	2.92
ACF of $\lvert r\rvert$, lag 1	0.27	0.00	0.00	0.28
ACF of $\lvert r\rvert$, sum of lags 1–20	5.32	0.00	0.00	4.79

- $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$
- Model columns: medians of 100 simulations
- A Heston model with a lower Feller ratio could reach the kurtosis
  - short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$
- Heston column: 390 Euler steps per day
  - with 780 steps, each median changes by less than 0.01

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Further Study: Reading

- Heavier tails: Lévy processes and α-stable distributions Cont & Tankov (2004), Pele (2014)
  - Crypto-assets form a distinct class by the shape of their return distributions Pele et al. (2023)
  - High-frequency information about Bitcoin volatility and tail risk Pele & Mazurencu-Marinescu-Pele (2019)
- Stochastic calculus
  - Shreve (2004), Ch. 5–6
  - Franke, Härdle & Hafner (2019), chapters on the Wiener process and stochastic integration
- Exercises with solutions on all models of this chapter: Borak, Härdle & López-Cabrera (2013)
  - the Heston model: Čížek, Härdle & Weron (2011)
  - applications: Härdle, Chen & Overbeck (2017)
- Monte Carlo methods: Glasserman (2003)
- Numerical SDEs: Kloeden & Platen (1992)

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Case Study, Model, Data and Method (1/4)

- **Model**: $\sigma_t = \xi e^{X_t}$ (eq. (1.3)), $X_t = \int_{-\infty}^{t} g(t-s)\, v_s \, \mathrm{d}W_s$ (eq. (1.5))
  - X is a Brownian semistationary (BSS) process with kernel g
  - v_s is the volatility-of-volatility process
  - the forecasts take $v_s \equiv 1$, a Gaussian X (Section 5)
- Gamma kernel $g(x) = x^{\alpha} e^{-\lambda x}$ (Example 1.2): roughness α, short memory
  - memory decays exponentially at rate λ
- Power kernel $g(x) = x^{\alpha}(1+x)^{-\gamma-\alpha}$ (eq. (1.8)): the Power-BSS model
  - $\beta = \gamma$ if $\gamma > 1$ (Remark 1.2)
  - $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$
- Gaussian alternative: the Cauchy class (Section 1.1.1)
  - $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$: roughness α and memory β set separately

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Case Study, Model, Data and Method (2/4)

- **Step 1**: α from the variogram of $\log \hat{\sigma}$, by OLS (eq. (2.1)) and NLLS
  - as in the SPY replication
  - the NLLS estimate averages $m = 10, \ldots, 20$ lags (Sections 2.1 and 3.1)
- **Step 2**: β, γ or λ by matching the model ACF to the empirical one
  - on the first $\lceil n^{1/3} \rceil$ lags (Sections 2.2 and 4.1)
- **Data**: E-mini S&P 500 futures, every trade, 2011–2014 (996 days)
  - trading window: 09:30–16:00 New York time
- Daily data on 1 944 liquid US stocks, 2003–2013 (Sections 4.1 and 4.2)

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Case Study, Model, Data and Method (3/4)

- $\hat{\sigma}_t^2$: pre-averaged bipower variation over Δ, divided by Δ
  - Δ from 10 minutes to 1 day (Sections 4 and 4.1)
- Bipower variation of the M returns r_j in a bin: $\mathrm{BV} = \frac{\pi}{2} \sum_{j=2}^{M} |r_j| \, |r_{j-1}|$
  - robust to jumps Barndorff-Nielsen & Shephard (2004)
- Pre-averaging first averages returns over short windows
  - aim: damp microstructure noise (Appendix A of the paper)
- Intraday seasonality removed with the flexible Fourier form of Andersen & Bollerslev (1997)

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Case Study, Model, Data and Method (4/4)

- **Forecasts**: variance summed over the next $h = 1, 2, 5, 10, 20$ steps
  - rolling window of 200 observations
  - out of sample: 2 January 2013 – 31 December 2014 (Section 5.1.1)
  - rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR Corsi (2009), ARFIMA(0,d,0), RFSV
  - log-HAR: heterogeneous autoregressive model for log volatility
- **Forecast object**: $\mathrm{FO}_t(\Delta, h) = \sum_{k=1}^{h} \mathrm{BV}_{t+k\Delta}$ (eq. (5.3))
  - Gaussian log model: $\mathbb{E}[e^{X_{t+\Delta}} \mid \mathcal{F}_t] = \exp(m + s^2/2)$ (eq. (5.2))
  - m, s^2: the conditional mean and variance
- Losses over the out-of-sample origins
  - $\mathrm{MSE} = \mathrm{mean}\,(\widehat{\mathrm{FO}}_t - \mathrm{FO}_t)^2$
  - $\mathrm{QLIKE} = \mathrm{mean}\,(\log \widehat{\mathrm{FO}}_t + \mathrm{FO}_t / \widehat{\mathrm{FO}}_t)$
- **Model confidence set** (MCS) Hansen, Lunde & Nason (2011): contains the best model with probability 90% (or 75%)
  - membership is not a ranking: the models inside the MCS cannot be told apart at that level
- Scaling BV by $M/(M-1)$ (Seminar 11, C3) shifts x_t by a constant
  - the shift: $\frac{1}{2}\ln(78/77) \approx 0.006$ for $M = 78$
  - the constant cancels in $\hat{\gamma}_2(k)$ and in the ACF

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Case Study, Results in the Paper (1/2)

$\log \sigma$, E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{\mathrm{OLS}}$	$\hat{\alpha}^*_{\mathrm{NLLS}}$	$\hat{\beta}^*_{\mathrm{Cauchy}}$	$\hat{\lambda}^*_{\mathrm{Gamma}}$
$\Delta = 10$ minutes	-0.38	-0.38	0.18	0.02
$\Delta = 65$ minutes	-0.30	-0.35	0.18	0.02
$\Delta = 1$ day	-0.30	-0.33	0.18	0.02

- Rough at every scale and persistent: long memory together with roughness
  - rough: $\hat{\alpha} < 0$, far rougher than Brownian motion ($\alpha = 0$)
  - persistent: $\hat{\beta}^* < 1$
  - * = noise-robust
  - SPY comparison in the main text

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Case Study, Results in the Paper (2/2)

- 1 944 US stocks: both features look universal (Figures 4 and 5)
  - $\hat{\alpha}^*$ mostly between -0.45 and -0.30
  - $\hat{\beta}^*$ mostly between 0 and 0.4
  - no systematic difference across sectors
- Noise can fake roughness (Table 1, Panel C)
  - simulation: $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$
  - the mean OLS estimate is -0.47
  - the mean NLLS estimate is 0.01
- Forecasts: modelling persistence pays intraday and at long horizons
  - 15-minute steps, $h = 20$, MSE ($\times 10^{8}$): 0.038 log-HAR, 0.020 RFSV, 0.019 Power-BSS (Table 4, Panel A)
  - daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^{9}$) (Table 5, Panel B)

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Case Study, Intraday Application, Limits and Open Questions (1/3)

- **Intraday application** on the SPY 5-minute bars of the course data
  - illustrative, not an exact replication
  - $\hat{\sigma}_t^2$ = bipower variation of the 5-minute returns in each bin of $\Delta = 30$ or 65 minutes
  - Divided by Δ and by the flexible Fourier seasonal factor Andersen & Bollerslev (1997)
- OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A
- log-HAR, RFSV and BSS forecasts against Table 5
  - losses compared: MSE, QLIKE and MCS
- Data limit: bars instead of every trade
  - the pre-averaged estimator (Appendix A of the paper) is not feasible
  - the noise is larger
  - only 13 returns per 65-minute bin

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Case Study, Intraday Application, Limits and Open Questions (2/3)

- **Limits stated by the authors**
  - Realised measures are noisy estimates of integrated variance
  - The conclusions concern them, not the latent spot volatility (Sections 4.1, 4.2 and 6)
  - No rule for the NLLS bandwidth m (Sections 2.1 and 3.1)
  - NLLS is too costly for rolling forecasts
  - These forecasts use the OLS $\hat{\alpha}$ (Section 5.1.1)
  - Forecasts assume a Gaussian X
  - A non-Gaussian volatility of volatility is left out of scope (Section 5; footnote 3)

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Case Study, Intraday Application, Limits and Open Questions (3/3)

- **Open question**: why does long memory improve even short-horizon forecasts?
  - The authors leave it to further research (Section 5.1.2)
  - Rolling $\hat{\alpha}$ is less rough in 2007–2008 and 2011 (Figure 3)
  - windows: 65 minutes, 2005–2014
  - median $\hat{\alpha}$: -0.36
  - $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6)
  - Does Bitcoin (5-minute bars), a market that never closes, fit this pattern?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot)
  - literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate estimates by frequency
  - check every DOI on Crossref (the title must match)
  - `Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.`
- **2. Competing hypotheses**, with distinct testable implications
  - `For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.`
- **3. Data and code**: replicate the chapter's estimate first
  - `Write Python that reads the S&P 500 daily closes of the course and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points).`

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The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered
  - `Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at t-1; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.`
- **5. Critique**: ask the assistant to attack the inference
  - `Act as a referee: why is the LR test of lambda = 0 non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?`
- **6. Report**: contents
  - all frequencies and filters
  - bootstrap critical values
  - code and the AI log

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Further Study on Quantinar

Advanced courses on quantinar.com

- Applied Computational Finance
- XFG Advanced Methods in Quantitative Finance

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

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Acronyms Used in This Material (1/4)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BSS**: Brownian SemiStationary (process)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CBOE**: Chicago Board Options Exchange
- **CI**: Confidence Interval
- **CIR**: Cox–Ingersoll–Ross (model)

Acronyms Used in This Material (2/4)

- **CKLS**: Chan–Karolyi–Longstaff–Sanders (model)
- **CLT**: Central Limit Theorem
- **CME**: Chicago Mercantile Exchange
- **DGS10**: 10-Year Treasury Constant Maturity Rate (FRED series code)
- **DOI**: Digital Object Identifier
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code)
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GBM**: Geometric Brownian Motion
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)

Acronyms Used in This Material (3/4)

- **JB**: Jarque–Bera (test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MCMC**: Markov Chain Monte Carlo
- **MCS**: Model Confidence Set
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **NLLS**: Non-Linear Least Squares
- **ODE**: Ordinary Differential Equation
- **OLS**: Ordinary Least Squares
- **OU**: Ornstein–Uhlenbeck (process)
- **PDE**: Partial Differential Equation

Acronyms Used in This Material (4/4)

- **QLIKE**: Quasi-LIKElihood (loss function)
- **QQ**: Quantile–Quantile (plot)
- **RFSV**: Rough Fractional Stochastic Volatility (model)
- **SDE**: Stochastic Differential Equation
- **SE**: Standard Error
- **SPY**: ticker of the SPDR S&P 500 ETF Trust
- **SVJ**: Stochastic Volatility with Jumps (model)
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VaR**: Value-at-Risk
- **VIX**: Cboe Volatility Index
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

References (1/5)

- Aït-Sahalia, Y. (1996). Testing continuous-time models of the spot interest rate. *Review of Financial Studies*, 9(2), 385–426.
- Aït-Sahalia, Y. (2002). Maximum likelihood estimation of discretely sampled diffusions: a closed-form approximation approach. *Econometrica*, 70(1), 223–262.
- Andersen, L. (2008). Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3), 1–42.
- Andersen, T.G., Benzoni, L., Lund, J. (2002). An empirical investigation of continuous-time equity return models. *Journal of Finance*, 57(3), 1239–1284.
- Andersen, T.G., Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3), 115–158.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bandi, F.M., Phillips, P.C.B. (2003). Fully nonparametric estimation of scalar diffusion models. *Econometrica*, 71(1), 241–283.
- Barndorff-Nielsen, O.E., Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.
- Bates, D.S. (1996). Jumps and stochastic volatility: exchange rate processes implicit in Deutsche Mark options. *Review of Financial Studies*, 9(1), 69–107.
- Bennedsen, M., Lunde, A., Pakkanen, M.S. (2022). Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5), 961–1006.
- Billingsley, P. (1999). Convergence of Probability Measures, 2nd ed. Wiley.
- Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Bollerslev, T., Zhou, H. (2002). Estimating stochastic volatility diffusion using conditional moments of integrated volatility. *Journal of Econometrics*, 109(1), 33–65.

References (2/5)

- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). Statistics of Financial Markets: Exercises and Solutions, 2nd ed. Springer.
- Broadie, M., Kaya, Ö. (2006). Exact simulation of stochastic volatility and other affine jump diffusion processes. *Operations Research*, 54(2), 217–231.
- Chan, K.C., Karolyi, G.A., Longstaff, F.A., Sanders, A.B. (1992). An empirical comparison of alternative models of the short-term interest rate. *Journal of Finance*, 47(3), 1209–1227.
- Chernov, M., Ghysels, E. (2000). A study towards a unified approach to the joint estimation of objective and risk neutral measures for the purpose of options valuation. *Journal of Financial Economics*, 56(3), 407–458.
- Christensen, K., Oomen, R.C.A., Podolskij, M. (2014). Fact or friction: jumps at ultra high frequency. *Journal of Financial Economics*, 114(3), 576–599.
- Christensen, K., Oomen, R.C.A., Renò, R. (2022). The drift burst hypothesis. *Journal of Econometrics*, 227(2), 461–497.
- Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). Statistical Tools for Finance and Insurance, 2nd ed. Springer.
- Comte, F., Renault, E. (1998). Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4), 291–323.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Cont, R., Tankov, P. (2004). Financial Modelling with Jump Processes. Chapman & Hall/CRC.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- Duffie, D., Pan, J., Singleton, K. (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, 68(6), 1343–1376.
- Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 322(8), 549–560.
- Eraker, B., Johannes, M., Polson, N. (2003). The impact of jumps in volatility and returns. *Journal of Finance*, 58(3), 1269–1300.

References (3/5)

- Feller, W. (1951). Two singular diffusion problems. *Annals of Mathematics*, 54(1), 173–182.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- Girsanov, I.V. (1960). On transforming a certain class of stochastic processes by absolutely continuous substitution of measures. *Theory of Probability and Its Applications*, 5(3), 285–301.
- Glasserman, P. (2003). Monte Carlo Methods in Financial Engineering. Springer.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). Applied Quantitative Finance, 3rd ed. Springer.
- Heston, S.L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.
- Higham, D.J. (2001). An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3), 525–546.
- Honoré, P. (1998). Pitfalls in estimating jump-diffusion models. Working paper, SSRN.
- Itô, K. (1944). Stochastic integral. *Proceedings of the Imperial Academy*, 20(8), 519–524.
- Kac, M. (1949). On distributions of certain Wiener functionals. *Transactions of the American Mathematical Society*, 65(1), 1–13.
- Kloeden, P.E., Platen, E. (1992). Numerical Solution of Stochastic Differential Equations. Springer.

References (4/5)

- Lee, S.S., Mykland, P.A. (2008). Jumps in financial markets: a new nonparametric test and jump dynamics. *Review of Financial Studies*, 21(6), 2535–2563.
- Lord, R., Koekkoek, R., van Dijk, D. (2010). A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2), 177–194.
- Merton, R.C. (1973). Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- Merton, R.C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- Milstein, G.N. (1975). Approximate integration of stochastic differential equations. *Theory of Probability and Its Applications*, 19(3), 557–562.
- Nelson, D.B. (1990). ARCH models as diffusion approximations. *Journal of Econometrics*, 45(1–2), 7–38.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Osborne, M.F.M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- Pele, D.T. (2014). A SAS approach for estimating the parameters of an alpha-stable distribution. *Procedia Economics and Finance*, 10, 68–77.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.

References (5/5)

- Pritsker, M. (1998). Nonparametric density estimation and tests of continuous time interest rate models. *Review of Financial Studies*, 11(3), 449–487.
- Samuelson, P.A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–39; reprinted in *Henry P. McKean Jr. Selecta* (2015), Birkhäuser.
- Shreve, S.E. (2004). Stochastic Calculus for Finance II: Continuous-Time Models. Springer.
- Stanton, R. (1997). A nonparametric model of term structure dynamics and the market price of interest rate risk. *Journal of Finance*, 52(5), 1973–2002.
- Tang, C.Y., Chen, S.X. (2009). Parameter estimation and bias correction for diffusion processes. *Journal of Econometrics*, 149(1), 65–81.
- Uhlenbeck, G.E., Ornstein, L.S. (1930). On the theory of the Brownian motion. *Physical Review*, 36(5), 823–841.
- Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.
- Wang, Y. (2002). Asymptotic nonequivalence of GARCH models and diffusions. *Annals of Statistics*, 30(3), 754–783.
- Wiener, N. (1923). Differential-space. *Journal of Mathematics and Physics*, 2(1–4), 131–174.
- Yu, J. (2012). Bias in the estimation of the mean reversion parameter in continuous time models. *Journal of Econometrics*, 169(1), 114–122.